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Do Sustainability Committees Mitigate or Exacerbate ESG Decoupling?

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ABSTRACT

This study investigates the impact of sustainability committees (SCs) on environmental, social, and governance (ESG) decoupling in US publicly listed firms. In particular, it examines their influence on overall and dimension-specific (E, S, G) ESG decoupling and distinguishes their effects on internal versus external ESG actions. The study also tests whether external monitoring factors, including analyst coverage, institutional ownership, and media scrutiny, moderate the relationship between SCs and ESG decoupling. Using a sample of 2759 firms with 20,038 firm-year observations from 2002 to 2021, we find that an SC significantly increases overall ESG decoupling, indicating that firms with SCs tend to disclose more external ESG claims without corresponding improvements in internal ESG practices. Dimension-specific analyses show that SCs reduce environmental decoupling but increase social and governance decoupling, highlighting heterogeneous effects across ESG dimensions. Moderation analysis shows that analyst coverage strengthens ESG decoupling, institutional ownership mitigates it, and media scrutiny exerts mixed effects across ESG dimensions. Overall, the findings suggest that SCs promote symbolic rather than substantive ESG practices, thus exacerbating ESG decoupling, particularly in social and governance domains.

1 | Introduction

Over the past two decades, environmental, social, and governance (ESG) considerations have become central to research on responsible corporate behavior. Firms are increasingly expected to align their strategies with broader societal goals, including climate responsibility, employee welfare, and ethical governance (Du and Vieira 2012). These developments reflect fundamental shifts in stakeholder expectations and investor priorities, as evidenced by the rise of ESG-driven investment strategies (Khan et al. 2016). According to Shmelev and Gilardi (2025), this shift signals growing societal demands for transparency, ethical accountability, and environmental stewardship. As a result, firms face mounting pressure to communicate their ESG commitments through voluntary and mandatory disclosures. Such disclosures, however, have also contributed to ESG decoupling, defined as the gap between a firm's ESG claims and

its actual practices, which can undermine stakeholder trust and distort market perceptions (Delmas and Burbano 2011; Lyon and Montgomery 2015).

ESG decoupling often appears through practices such as greenwashing or bluewashing, in which firms highlight positive ESG outcomes while minimizing or omitting negative aspects of their operations, thereby misleading stakeholders about actual sustainability performance (Lyon and Maxwell 2011). Decoupling is prevalent in voluntary disclosure environments, such as the United States, where ESG reporting lacks standardization and rigorous third-party verification (Marquis et al. 2016).

Firms face several challenges in aligning internal ESG actions with external reporting. Integrating sustainability practices requires operational changes, financial resources, and specialized expertise (Thun et al. 2024). Stakeholders often evaluate

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firms based on disclosed ESG information rather than actual practices, encouraging symbolic communication (Sulemana et al. 2025). External pressures from stakeholders create strong incentives for firms to emphasize visible sustainability credentials to maintain legitimacy (Alessa et al. 2024). Heightened information asymmetry further enables selective reporting, exaggeration of positive outcomes, or omission of negative information (Ivić and Cerić 2023). Moreover, firms must navigate the complexity of integrating ESG practices across operational, strategic, and governance levels to ensure that internal actions align with external disclosures. Although frameworks such as the Global Reporting Initiative and the Sustainability Accounting Standards Board offer guidance, firms still retain considerable discretion over what, how, and when they report ESG information, thus enabling symbolic reporting (Ilori et al. 2023). These challenges highlight the risk of ESG decoupling and underscore the importance of governance mechanisms that encourage authentic sustainability.

To address these issues, many firms have established sustainability committees (SCs), which are board-level committees responsible for overseeing ESG strategies. These committees are intended to strengthen accountability, guide ESG initiatives, and enhance transparency (Orazalin 2020) and can improve environmental performance and disclosure quality (Driss et al. 2024; Alodat et al. 2025). Despite their growing prevalence, however, gaps remain concerning their overall effectiveness and potential unintended consequences.

Although existing studies focus primarily on the positive effects of SCs, much less is known about the unintended effects related to ESG decoupling. Prior research has not examined whether SCs help narrow the gap between internal ESG practices and external disclosures, or whether they inadvertently encourage symbolic reporting, particularly in the social and governance domains (Gull et al. 2023). This study addresses this gap by examining the impact of SCs on overall and dimension-specific (E, S, and G) ESG decoupling. Accordingly, our first research question is as follows:

1. *Do SCs reduce or exacerbate ESG decoupling and are these effects consistent across ESG dimensions?*

Many studies treat ESG activities as a unified construct rather than distinguishing between internal ESG practices, which reflect substantive actions, and external ESG disclosures, which reflect symbolic communication (Shah et al. 2024). This limitation makes it difficult to determine whether SCs genuinely improve sustainability performance or primarily influence outward-facing reporting. Distinguishing between internal and external ESG actions is essential because firms may engage in extensive external disclosures without corresponding internal improvements. Our second research question is as follows:

2. *Do SCs contribute more to substantive internal ESG improvements or to symbolic external disclosures that widen the ESG decoupling gap?*

Finally, external stakeholders, such as analysts, institutional investors, and the media, play a critical role in monitoring corporate behavior and influencing ESG practices. These stakeholders

can pressure firms to improve transparency and better align internal ESG practices with external disclosures. Greater analyst coverage may increase reputational risks for firms that engage in symbolic ESG reporting (M. Zhang et al. 2015), whereas media scrutiny can elevate public attention and discourage misleading practices (Cahan et al. 2015). Institutional investors may also monitor sustainability actions, reducing the likelihood of ESG decoupling (Li et al. 2021). Nevertheless, the moderating role of these stakeholders in shaping the relationship between SCs and ESG decoupling remains unclear. It is not yet known whether SCs are more effective in reducing ESG decoupling under high external scrutiny or whether such oversight may unintentionally encourage symbolic reporting to meet stakeholder expectations. This study examines the moderating effects of analyst coverage, institutional ownership, and media scrutiny on the relationship between SCs and ESG decoupling. Accordingly, the third research question is as follows:

3. *To what extent do external stakeholders moderate the relationship between SCs and ESG decoupling in US publicly listed firms?*

Using a panel dataset of 2759 US publicly listed firms with 20,038 firm-year observations from 2002 to 2021, we find that SCs increase overall ESG decoupling by promoting symbolic, externally visible ESG actions without corresponding internal improvements. Although SCs reduce environmental decoupling through substantive internal and external environmental initiatives, they enhance external social and governance actions, thus increasing decoupling in these dimensions. These effects are moderated by external monitoring factors: analyst coverage amplifies decoupling in the social and governance dimensions, media scrutiny has mixed effects across ESG dimensions, and institutional ownership generally mitigates decoupling. Overall, our results indicate that SCs tend to foster legitimacy-driven ESG engagement rather than substantive transformation, with heterogeneous effects across the ESG dimensions.

This study contributes to the literature on business ethics, ESG governance, and sustainability reporting in several ways. First, it provides a comprehensive analysis of the dual role of SCs in shaping ESG practices. Unlike prior research that focuses mainly on positive ESG outcomes, this study shows that SCs may also encourage ESG decoupling. To our knowledge, SCs have not been systematically examined in relation to ESG decoupling, nor has their predictive role been evaluated in a market-oriented economy such as the United States, where ESG reporting remains largely voluntary and unstandardized.

Second, we document dimension-specific effects, showing that SCs reduce environmental decoupling but increase social and governance decoupling. Third, by distinguishing between internal ESG practices and external ESG disclosures, we show that SCs tend to promote visible external initiatives rather than substantive internal improvements. Fourth, by examining how external monitoring by analysts, institutional investors, and the media moderates the effect of SCs, we highlight the conditional role of external stakeholders in constraining or enabling symbolic sustainability behavior. These findings deepen understanding of when and how board governance mechanisms support authentic sustainability efforts versus symbolic ESG reporting.

Finally, the study clarifies how internal governance and external monitoring interact to shape the balance between symbolic and substantive ESG practices, offering implications for policy-makers, corporate boards, ESG rating agencies, and stakeholders seeking to align disclosures with actual performance.

The remainder of this paper is organized as follows. Section 2 reviews prior literature on SCs, ESG decoupling, and external monitoring factors. Section 3 describes the methodology and data collection process. Section 4 presents and discusses the results. Section 5 concludes the study and outlines its practical implications.

2 | Literature Review

2.1 | Background on SCs

SCs have emerged as important board-level governance mechanisms that oversee a firm's ESG strategies, guide sustainability initiatives, and enhance transparency and accountability in ESG reporting (Miller and Serafeim 2014). As stakeholder expectations regarding ESG issues have intensified, many firms have restructured their traditional governance structures by establishing dedicated board-level committees to oversee sustainability and stakeholder-related concerns. These committees are typically voluntary and operate alongside mandatory committees, such as audit, compensation, and nomination committees. According to Alcaide-Ruiz et al. (2022), an SC is a board-level committee or subcommittee within a firm's governance structure that assists in developing and implementing sustainable strategies, overseeing sustainability reporting, promoting environmental and social innovation, and monitoring climate- and sustainability-related risks. Similarly, the Board Portal glossary defines an SC as a board-constituted body responsible for "overseeing, developing, and promoting sustainable practices and initiatives," ensuring that company operations remain environmentally, socially, and economically responsible to stakeholders. Burke et al. (2019) defined SCs as an extension of corporate governance that oversees the firm's impacts on multiple stakeholder groups, including the community, employees, the environment, consumers, and suppliers. This definition clarifies the strategic and stakeholder-oriented role of SCs. The establishment of SCs reflects firms' recognition of sustainability as a strategic imperative and a response to increasing stakeholder and societal expectations of corporate responsibility.

Importantly, SCs are conceptually distinct from CSR committees. Although CSR committees are often associated with philanthropic initiatives, social programs, and disclosure practices, SCs typically adopt a broader and more strategic role by integrating ESG considerations into core business decision-making, overseeing sustainability risks and opportunities, and aligning ESG objectives with corporate strategy.

However, a key characteristic of SCs is their heterogeneity across firms and industries. SCs do not have uniform governance structures, and their mandates, composition, and focus vary depending on firm-specific characteristics, industry exposure, and stakeholder pressures (Burke et al. 2019). For example, firms operating in environmentally sensitive industries often emphasize

environmental risk management and regulatory compliance, whereas firms in other sectors may prioritize social, ethical, or innovation-related concerns. For instance, Shell emphasizes environmental and social risk management within its sustainability governance structures, whereas AstraZeneca complements sustainability oversight with a strong focus on scientific governance and research integrity. Similarly, Land Securities integrates sustainability governance with environmental and energy strategy. These examples demonstrate that SCs reflect firm-specific strategic priorities rather than a homogeneous governance model, reinforcing the importance of considering their functional diversity.

Empirical research highlights several functions and benefits of SCs, including improving ESG performance (Hussain et al. 2018), enhancing the quality and credibility of sustainability disclosures (Dixon-Fowler et al. 2017), integrating ESG considerations into corporate strategy (Li et al. 2025), and monitoring climate- and sustainability-related risks (Alcaide-Ruiz et al. 2022). SCs also strengthen risk management frameworks by identifying and mitigating environmental and social risks that may affect long-term firm value (Agustina et al. 2016). Furthermore, SCs can enhance legitimacy and reputation, as establishing a dedicated committee signals a firm's commitment to transparency and long-term sustainability (Michelon and Parbonetti 2012; Peters and Romi 2015). SCs may also facilitate stakeholder engagement by ensuring that corporate ESG initiatives respond to the expectations of investors, regulators, and broader society (Miller and Serafeim 2014). By institutionalizing oversight at the board level, SCs provide a structured mechanism to integrate sustainability into corporate strategy, manage ESG-related risks, and align corporate actions with stakeholder interests.

However, although most studies emphasize their positive role, SCs may also facilitate symbolic ESG governance, in which firms enhance sustainability disclosures without corresponding improvements in actual ESG performance, creating a divergence between communication and practice (Rodrigue et al. 2013). The effectiveness of SCs in promoting substantive ESG improvements depends on several interrelated factors, including the responsibilities outlined in their charters, which determine their authority to oversee ESG strategy and monitor sustainability performance (Alcaide-Ruiz et al. 2022); the sustainability expertise of committee members, such as knowledge of environmental management, social responsibility, governance practices, and sustainability accounting (Haque 2017; Martín and Herrero 2020); and their composition, including size, independence, diversity, and tenure, which influence decision quality and oversight capacity (Rodriguez-Fernandez 2016). Operational characteristics, such as meeting frequency, coordination with other board committees, and the robustness of reporting processes, also shape effectiveness by influencing oversight intensity and information flow (Peters and Romi 2015). The broader industry context plays an additional role, as firms in environmentally sensitive or high-impact industries face stronger stakeholder scrutiny and regulatory pressures, affecting how SCs integrate ESG considerations into strategy and mitigate decoupling (Marquis et al. 2016). Committees with members who possess relevant sustainability expertise, clear mandates, and sufficient decision-making authority are more likely to drive

substantive ESG outcomes. In contrast, committees established primarily for symbolic purposes may contribute to selective or superficial reporting. A comprehensive understanding of the role, structure, and functioning of SCs is therefore essential for assessing their true impact on corporate sustainability practices and disclosures and for identifying the conditions under which they promote authentic ESG improvements rather than merely enhancing a firm's sustainability image.

2.2 | Determinants of SC Formation

The formation of SCs represents an important strategic governance decision that reflects how firms internalize external sustainability pressures and integrate them into corporate structures. Both internal firm characteristics and external pressures influence the likelihood of establishing such committees (McCullough et al. 2018). Internal determinants include firm size, financial resources, board composition, managerial expertise, and prior ESG performance, all of which shape a firm's capacity and willingness to institutionalize sustainability oversight. Firm size is a key factor, and Bolibok (2024) argues that larger firms possess greater human, capital, and organizational resources, enabling them to manage ESG risks more effectively and implement structured oversight mechanisms. Financial resources are also important, as firms with sufficient organizational slack are better positioned to support the costs associated with board-level SCs, including reporting, risk monitoring, and ESG-related initiatives (Heubeck and Ahrens 2025).

Board composition is another important determinant. Board independence, diversity, and prior expertise influence the quality of oversight provided by SCs. J. Q. Zhang et al. (2013) find that more independent and diverse boards are better equipped to form effective committees capable of overseeing ESG strategies. Managerial expertise also affects the decision to form SCs. Firms with leaders who possess experience in environmental management, social responsibility, governance practices, or sustainability reporting are more likely to recognize the strategic importance of formal ESG oversight and advocate for committee formation. For example, Martín and Herrero (2020) show that firms with boards possessing environmental expertise are more likely to adopt formal governance mechanisms, including SCs. Collevocchio et al. (2025) find that board members with sustainability experience are especially influential when combined with complementary characteristics such as board age and tenure. Prior ESG performance may also drive committee formation. Firms with strong records may establish SCs to formalize and strengthen existing governance systems, ensuring that sustainability principles are embedded in strategic decision-making (Rodriguez-Fernandez 2016). Conversely, firms with weaker ESG performance may introduce such committees in response to legitimacy pressures, aiming to improve accountability, signal commitment to responsible practices, and rebuild stakeholder trust (Peters and Romi 2015). Thus, both high- and low-ESG performers may form SCs, although their motivations differ.

External pressures also play a significant role in the establishment of SCs. Regulatory expectations can motivate firms to institutionalize sustainability practices to comply with evolving

disclosure requirements and environmental reporting standards (Liao et al. 2015). Investor scrutiny reinforces this effect, as institutional investors increasingly demand transparent governance structures to mitigate long-term risks and enhance accountability (Dyck et al. 2019). Media attention heightens reputational risks, prompting firms to adopt SCs as a signal of responsible behavior to mitigate investor and stakeholder concerns (Q. Zhang and Wong 2022). Stakeholder demands from customers, employees, NGOs, and local communities similarly encourage firms to integrate sustainability at the board level to maintain legitimacy and respond to rising societal expectations (Grewatsch and Kleindienst 2017; Lee 2011). These internal and external determinants shape not only whether firms form SCs but also their mandates, responsibilities, and effectiveness. Firms that align internal capabilities with external pressures are more likely to develop committees that promote substantive ESG outcomes, whereas firms lacking such alignment may establish committees that contribute primarily to symbolic or superficial reporting.

2.3 | ESG Decoupling

ESG decoupling refers to the divergence between a firm's internal ESG practices and its external ESG disclosures (Yamagata 2025; Eliwa and Elmaghrabi 2025). Firms may undertake substantive ESG activities internally, whereas their external communications exaggerate achievements or selectively report favorable outcomes, giving stakeholders a misleading impression of actual sustainability performance (Tian and Niu 2024). This phenomenon often manifests through greenwashing or bluewashing, where firms highlight positive ESG outcomes while minimizing or omitting negative aspects of their operations (Roszkowska-Menkes et al. 2024; Eliwa et al. 2025).

Several internal and external factors contribute to ESG decoupling. Internal drivers include weak governance structures, a lack of board-level oversight, and limited managerial expertise. For example, Boiral (2016) shows that firms with weak monitoring systems often use biodiversity reporting to manage stakeholder perceptions rather than communicate substantive environmental actions. Such firms may selectively disclose favorable information, use ambiguous language, or emphasize minor initiatives, thereby widening the gap between reported ESG commitments and actual practices. A lack of board-level oversight similarly reduces monitoring of ESG performance (Delmas and Burbano 2011). Limited managerial expertise can also exacerbate decoupling. Liu and Zhang (2023) find that managerial myopia and insufficient ESG knowledge undermine firms' ability to design, implement, and track effective ESG strategies, increasing the misalignment between ESG performance and external disclosures.

External factors also shape ESG decoupling. Voluntary disclosure regimes, limited regulatory enforcement, and the absence of standardized reporting frameworks provide firms with considerable discretion in ESG reporting. Under voluntary disclosure systems, firms decide what, when, and how to report ESG information, which encourages selective disclosure and omission of negative data to maintain legitimacy and attract investors (Cho et al. 2015). Weak regulatory enforcement amplifies

this problem because the absence of penalties for misleading sustainability disclosures enables firms to engage in symbolic compliance or greenwashing without significant consequences (Lyon and Montgomery 2015). In addition, inconsistent or non-standardized reporting frameworks across countries and sectors result in incomparable disclosures, enabling firms to frame ESG narratives strategically rather than substantively (Ioannou and Serafeim 2019; Christensen et al. 2021).

ESG decoupling has serious consequences for firms, investors, and broader society. At the firm level, inconsistencies between reported and actual ESG practices can undermine stakeholder trust and damage corporate reputation once detected (Berg et al. 2022). This loss of credibility may reduce investor confidence (Di and Li 2023), restrict access to sustainable financing (Cepêda et al. 2025), and diminish firm value (Aouadi and Marsat 2018). For investors, ESG decoupling increases information asymmetry, distorts capital allocation, and may direct investment toward firms that appear sustainable on paper but fail to deliver substantive ESG outcomes (Yamagata 2025). At a broader societal level, widespread decoupling undermines the integrity of sustainability reporting, weakens market discipline, and reduces the overall effectiveness of ESG initiatives in addressing environmental and social challenges (Roszkowska-Menkes et al. 2024). Strengthening governance mechanisms, implementing mandatory disclosure standards, and enhancing regulatory oversight are therefore essential to ensure transparency, accountability, and long-term sustainability outcomes.

Given the drivers and consequences of ESG decoupling, it is critical to examine governance mechanisms that may mitigate or exacerbate this divergence. SCs are an important governance tool for overseeing ESG strategies, monitoring internal practices, and guiding external disclosures. The next section examines how these committees influence ESG decoupling and considers whether they reduce or intensify this divergence.

2.4 | SCs and ESG Decoupling

The role of SCs in shaping the alignment between ESG disclosure and actual performance remains theoretically and empirically contested. Rather than exerting a uniform effect, SCs operate under competing governance and signaling incentives, suggesting that their impact on ESG decoupling is inherently conditional and dimension-specific. Prior literature suggests that SCs can either mitigate or exacerbate ESG decoupling, depending on whether they function as substantive governance mechanisms or symbolic legitimacy tools.

From an agency-theoretic perspective, SCs are considered key mechanisms for enhancing board monitoring by improving oversight, strengthening internal controls, and reducing information asymmetry between managers and stakeholders (Driss et al. 2024). This monitoring function may limit opportunistic disclosure behavior and promote greater alignment between ESG disclosure and substantive performance. Empirical evidence suggests that SCs help mitigate ESG decoupling by strengthening the alignment between a firm's internal ESG practices and its external disclosures. By facilitating cross-functional coordination, strengthening internal controls, and promoting rigorous

board oversight, SCs can enhance transparency, accountability, and disclosure quality, thus increasing the credibility of ESG communication and narrowing the gap between reported and actual ESG outcomes (López-Arceiz et al. 2022; Abdullah et al. 2024).

In this sense, SCs act as internal governance mechanisms that embed ESG considerations into strategic decision-making and ensure consistent implementation of sustainability practices across the organization. Ludwig and Sassen (2022) argue that SCs are particularly effective in industries with significant environmental and social impacts, where they help reduce symbolic reporting, promote transparent disclosure, and support the effective implementation of ESG initiatives throughout the organization. For instance, García-Sánchez et al. (2022) note that committees with clear mandates, formal reporting structures, and direct access to the board are better positioned to integrate ESG considerations into corporate decision-making. Similarly, Velte (2017) finds that well-structured SCs enhance disclosure quality, effectively reducing the gap between internal ESG practices and external reports.

Firms with active and independent SCs are also more likely to integrate ESG principles into their strategies and operations. Alodat et al. (2025) find that SCs strengthen the relationship between ESG performance and environmental innovation by encouraging cross-functional collaboration and resource allocation for sustainable projects. Sahu et al. (2025) show that committee expertise in auditing and sustainability improves ESG quality by reinforcing internal controls and reducing greenwashing tendencies. Abdullah et al. (2024) note that SCs facilitate coordination across departments, helping firms achieve ESG targets and disclose information that accurately reflects performance. Rosyidah and Ningsih (2025) demonstrate that ESG committees improve corporate sustainability performance by enhancing governance structures, embedding sustainability considerations into strategy, and ensuring rigorous monitoring of ESG practices. Their findings suggest that ESG committees not only improve the quality and credibility of sustainability disclosures but also support genuine organizational transformation toward long-term sustainability. Collectively, these studies indicate that SCs, when characterized by independence, expertise, and board-level authority, can function as effective monitoring mechanisms that reduce ESG decoupling.

However, this governance effectiveness is unlikely to be uniform across ESG dimensions, as the nature of performance measurement and verification differs substantially between ESG domains. Environmental metrics are typically standardized, quantitative, and externally verifiable, such as carbon emissions, energy consumption, and pollution levels, which facilitates objective measurement and comparability (Tsang et al. 2023). These environmental indicators are increasingly integrated into formal ESG reporting frameworks, such as the Global Reporting Initiative, which enhances the reliability, comparability, and consistency of disclosure (Tsang et al. 2023). The higher measurability and verifiability of environmental performance strengthen the monitoring capacity of SCs, enabling them to more effectively enforce alignment between disclosed and actual environmental outcomes and thus reduce environmental decoupling. Empirical evidence

supports this view, showing that SCs are positively associated with higher-quality environmental disclosures (Driss et al. 2024).

In contrast, the social and governance dimensions are often more qualitative, contextual, and less standardized, involving social aspects (e.g., employee well-being, diversity, and community engagement), as well as governance aspects, such as board effectiveness or ethical culture. The lack of widely accepted standards and clear quantification criteria creates interpretive ambiguity and increases managerial discretion in how information is presented and framed in disclosures, particularly for social and governance performance (Jagrič et al. 2025; Fometescu et al. 2024). Such ambiguity makes it harder to objectively verify substantive performance, thus weakening the effectiveness of SCs and increasing the risk that social and governance disclosures reflect symbolic narratives rather than substantive change (Z. Zhang et al. 2026).

However, other researchers argue that SCs may instead exacerbate ESG decoupling, particularly when they exist only symbolically or operate under weak governance conditions. From a legitimacy theory perspective, SCs may be adopted primarily to manage stakeholder perceptions and signal commitment to sustainability rather than to drive substantive organizational change. In this context, SCs function as impression-management mechanisms that enhance perceived legitimacy while allowing firms to maintain existing practices. Under voluntary disclosure regimes, firms have significant discretion in deciding what, when, and how to report ESG information (Cho et al. 2015), thus increasing the risk of ESG decoupling. Directors with ESG knowledge may use that expertise to strategically manipulate disclosures by selecting which information to emphasize, omit, or frame positively to maintain legitimacy and attract investors (Konstantin and Elena 2022). Yamagata (2025) similarly notes that some firms establish SCs primarily to satisfy stakeholder expectations rather than to improve ESG performance. These committees may focus more on managing external perceptions than on driving internal sustainability improvements. Khamisu et al. (2024) find that firms in environmentally sensitive industries often disclose ESG information mainly for legitimacy and reputation management rather than substantive environmental or social improvements.

This symbolic use of SCs is particularly pronounced in the social and governance dimensions, where performance is inherently more difficult to quantify and more susceptible to narrative framing. Firms may emphasize symbolic initiatives, such as diversity policies or governance reforms, without corresponding substantive changes in underlying practices, thus widening the gap between disclosure and actual performance. Firms often engage in impression management and symbolic disclosure practices, emphasizing ESG initiatives without corresponding substantive changes in underlying activities (Cho et al. 2012; Boiral 2013), contributing to ESG decoupling (Hawn and Ioannou 2016). The lack of standardized and quantifiable metrics in the social and governance domains further facilitates selective reporting and increases the risk of ESG decoupling.

SCs may also be influenced by managerial or reputational incentives, leading to symbolic reporting rather than substantive

accountability. This risk is amplified when committees lack independence, possess limited ESG expertise, or have weak authority within the board structure, reducing their ability to challenge managerial opportunism. When committees lack independence or are dominated by executives with short-term financial goals, they may reinforce ESG decoupling by validating overly optimistic disclosures to satisfy investors and external rating agencies (Albitar et al. 2023; Roszkowska-Menkes et al. 2024). Weak enforcement mechanisms and fragmented reporting standards further enable these committees to selectively frame ESG disclosures to maintain stakeholder confidence, even when actual performance does not improve.

Taken together, the literature reveals a fundamental tension regarding the role of SCs in ESG decoupling. Although the agency perspective emphasizes the monitoring and governance benefits of SCs in reducing decoupling, the legitimacy perspective highlights their potential symbolic role in exacerbating it. This tension implies that the effect of SCs is asymmetric across ESG dimensions, driven by differences in measurability, verifiability, and managerial discretion.

Given increasing stakeholder and institutional pressure on firms to demonstrate their sustainability commitment, it is plausible that SCs are frequently adopted as signaling mechanisms to enhance legitimacy rather than as purely substantive governance tools. Consequently, although SCs may reduce environmental decoupling by increasing measurability and monitoring, they are more likely to increase social and governance decoupling where ambiguity and discretion dominate. At the aggregate level, these opposing effects may result in an overall increase in ESG decoupling.

Thus, the impact of SCs on ESG decoupling largely depends on their composition, independence, authority, and the broader institutional context in which they operate. Based on a substantive review of the literature, we hypothesize the following relationships:

Hypothesis 1. *SCs have a significant positive influence on overall ESG decoupling.*

Hypothesis 1a. *SCs have a significant negative influence on environmental decoupling.*

Hypothesis 1b. *SCs have a significant positive influence on social decoupling.*

Hypothesis 1c. *SCs have a significant positive influence on governance decoupling.*

2.5 | SCs and Their Impact on Internal and External ESG Actions

SCs play an increasingly important role in shaping corporate ESG activities, yet their influence differs markedly between internal and external ESG actions. Internal ESG actions refer to substantive and operational practices such as reducing emissions, improving labor conditions, strengthening governance controls, and integrating sustainability into core business

processes (de Souza Barbosa et al. 2023). External ESG actions involve outward-facing communication, including sustainability reporting, disclosure strategies, stakeholder engagement, and other activities aimed at shaping public perceptions of corporate responsibility (Tumewang et al. 2025).

SCs with relevant expertise and board authority can enhance internal ESG actions by coordinating resources, monitoring progress, and embedding sustainability into strategic and operational processes (Abdullah et al. 2024). This effect is expected to be strongest for environmental actions, where metrics are quantitative, verifiable, and integrated into reporting standards, allowing SCs to guide operational improvements and reinforce the alignment between internal practices and external expectations (Tsang et al. 2023).

For the social and governance dimensions, internal ESG actions are often qualitative, context-dependent, and less standardized. Consequently, SCs may have a limited capacity to enforce substantive internal changes, as measurement ambiguity and managerial discretion can weaken their effectiveness (Jagrič et al. 2025).

However, SCs may place disproportionate emphasis on external ESG actions, potentially increasing the risk of symbolic sustainability behaviors. Because SCs are highly visible governance structures, firms may use them strategically to signal their commitment to responsible practices, particularly under strong stakeholder or legitimacy pressure (Rodrigue et al. 2013; Yamagata 2025). In these situations, SCs may focus more on improving sustainability disclosure, enhancing ESG ratings, and managing external perceptions than on driving substantive internal changes. As SCs operate at the strategic level and are sensitive to stakeholder expectations, they may prioritize external ESG actions that are easier to implement and yield more immediate legitimacy benefits.

This symbolic emphasis likely varies across ESG dimensions. For environmental actions, SCs may still support substantive internal improvements while managing external disclosure (García-Sánchez et al. 2022). In contrast, social and governance actions are more qualitative, contextual, and less standardized, giving managers greater discretion to focus on external reporting rather than on operational change (Konstantin and Elena 2022). Therefore, the influence of SCs on external social and governance actions is likely stronger than on internal practices, increasing the risk of ESG decoupling in these dimensions (Khamisu et al. 2024). This dimension-specific perspective highlights why firms may publicly report strong social and governance performance even when internal improvements are limited, whereas environmental actions remain more closely aligned with actual operations (Abdullah et al. 2024).

Together, these insights suggest that the influence of SCs on ESG actions is both action- and dimension-specific. SCs may drive substantive internal improvements primarily for environmental actions, whereas their impact on social and governance practices may be more symbolic in nature. Conversely, SCs are likely to exert a stronger influence on external ESG actions across all dimensions, with social and governance actions

being particularly susceptible to signaling effects. This provides a clear theoretical basis for the following hypotheses:

Hypothesis 2. *SCs have a significant positive influence on external ESG actions.*

Hypothesis 2a. *SCs have a significant positive influence on external environmental actions.*

Hypothesis 2b. *SCs have a significant positive influence on external social actions.*

Hypothesis 2c. *SCs have a significant positive influence on external governance actions.*

Hypothesis 3. *SCs have a weak or negative influence on overall internal ESG actions.*

Hypothesis 3a. *SCs have a significant positive influence on internal environmental actions.*

Hypothesis 3b. *SCs have a significant negative influence on internal social actions.*

Hypothesis 3c. *SCs have a significant negative influence on internal governance actions.*

2.6 | Moderating Role of External Stakeholders

Although SCs play an important role in shaping firms' ESG actions, their effectiveness and the credibility of ESG outcomes often depend on external pressures. According to stakeholder theory (Freeman 1984) and legitimacy theory (Suchman 1995), corporate sustainability practices are shaped not only by internal motivations but also by external monitoring, expectations, and demands. External actors, particularly financial analysts, institutional investors, and the media, serve as influential information intermediaries that can either discipline firms toward genuine ESG improvements or motivate symbolic compliance aimed at satisfying stakeholder expectations. This subsection reviews the literature on how analyst coverage, institutional ownership, and media scrutiny moderate the relationship between SCs and ESG decoupling.

2.6.1 | Analyst Coverage

Analyst coverage reflects the extent to which a firm is monitored and evaluated by financial analysts who issue reports and recommendations to investors (Yu 2008). As an external governance mechanism, analyst coverage not only directly influences corporate behavior but also shapes the effectiveness of internal governance structures, such as SCs, in influencing ESG outcomes.

From an agency perspective, analyst coverage enhances external monitoring and reduces information asymmetry, which could strengthen the governance role of SCs and improve alignment between ESG disclosure and substantive performance. However, this monitoring role is accompanied by strong capital

market pressures that may alter managerial incentives and the functioning of SCs.

When firms attract extensive analyst attention, managers face intensified pressure to meet short-term earnings expectations and maintain favorable investor sentiment (Huang et al. 2017). Under such conditions, SCs may shift their focus toward external ESG disclosures that enhance firm's reputation among analysts and investors (Driss et al. 2024) while allocating fewer resources to substantive internal ESG improvements that generate long-term value but have limited immediate visibility.

From a signaling and legitimacy perspective, analyst coverage amplifies firms' incentives to strategically manage ESG disclosures. Under high analyst scrutiny, firms are more likely to use SCs as signaling devices to project strong ESG performance, even when internal practices do not fully align (Männistö 2023). Because analysts predominantly rely on publicly disclosed ESG information, rather than directly verifying underlying practices, this creates conditions under which symbolic ESG reporting can persist (Darnall et al. 2022). Therefore, SCs may inadvertently reinforce ESG decoupling under high analyst scrutiny, functioning as tools of impression management rather than as drivers of genuine ESG improvements.

Importantly, the moderating effect of analyst coverage is likely to vary across ESG dimensions. In environmental domains, analyst scrutiny may reinforce the substantive role of SCs and limit decoupling. In contrast, in social and governance dimensions, where performance is more qualitative and subject to managerial discretion, analyst pressure is more likely to increase the symbolic use of SCs, thereby increasing ESG decoupling.

Empirical evidence supports this view. Wang, Hu, and Wang et al. (2024) find that firms under intensive analyst scrutiny are more likely to engage in symbolic ESG disclosures rather than substantive ESG practices. Similarly, Dyck et al. (2019) show that analyst coverage strengthens reported ESG performance without corresponding improvements in underlying ESG outcomes. Therefore, higher analyst coverage may increase ESG decoupling by encouraging SCs to focus on reputation management and external disclosure rather than on substantive internal ESG improvements.

Overall, the evidence implies that analyst coverage modifies the relationship between SCs and ESG decoupling. Although SCs may function as governance mechanisms under low external pressure, high analyst coverage may shift their role toward impression management, thereby strengthening ESG decoupling, particularly in dimensions characterized by higher measurement ambiguity. This provides a theoretical basis for the following hypotheses:

Hypothesis 4. *Analyst coverage positively moderates the relationship between SCs and ESG decoupling, such that the positive association between SCs and ESG decoupling is stronger when analyst coverage is high.*

Hypothesis 4a. *Analyst coverage negatively moderates the relationship between SCs and environmental decoupling, such that the association is weaker when analyst coverage is high.*

Hypothesis 4b. *Analyst coverage positively moderates the relationship between SCs and social decoupling, such that the positive association is stronger when analyst coverage is high.*

Hypothesis 4c. *Analyst coverage positively moderates the relationship between SCs and governance decoupling, such that the positive association is stronger when analyst coverage is high.*

2.6.2 | Media Scrutiny

Media scrutiny refers to the extent to which firms are observed, evaluated, and publicized by mass media outlets, including newspapers, online platforms, and social media channels that disseminate information to the public and investors (Dyck and Zingales 2002). As an external governance mechanism, media scrutiny not only increases transparency but also shapes how internal governance structures, such as SCs, influence ESG performance.

The media can function as an important external monitoring force by disseminating information, shaping public perception, and influencing corporate behavior. Through its watchdog role, media exposure increases the visibility of firms' environmental and social practices and heightens accountability pressure on management (J. Luo et al. 2012).

From a legitimacy perspective, firms under intense media scrutiny face heightened pressure to conform to societal expectations and avoid reputation damage (Patten 2002). In response, SCs may prioritize visible ESG initiatives and disclosures to maintain legitimacy, even when such actions are not supported by substantive internal improvements.

From a signaling perspective, media scrutiny amplifies the incentives for firms to strategically communicate positive ESG information to external stakeholders (Connelly et al. 2011). Because media coverage shapes public narratives and investor perceptions, firms may rely on SCs to enhance sustainability disclosure, selectively highlight achievements, and manage impressions, particularly when facing negative publicity.

Under intensive media scrutiny, firms face reputational risks associated with negative coverage of unsustainable behavior or inconsistencies between ESG disclosures and actual practices (Ali and Ali 2025). To manage this risk, SCs may intensify symbolic ESG activities such as producing polished sustainability reports or selectively highlighting achievements (Levonian 2022). Consequently, media pressure may shift the role of SCs from substantive governance mechanisms to tools of impression management, reinforcing ESG decoupling.

Importantly, the moderating effect of media scrutiny varies across ESG dimensions. For environmental performance, media scrutiny may partially constrain symbolic behavior and limit decoupling. In contrast, the social and governance dimensions are more qualitative and narrative-driven, making them more susceptible to media framing and impression management. Therefore, media scrutiny is more likely to amplify the symbolic use of SCs in the social and governance domains, increasing ESG decoupling in these dimensions.

Empirical research supports these claims. Yang et al. (2024) find that firms exposed to media scrutiny engage in symbolic ESG reporting as a strategy to manage public perception. Luo and Tang (2014) show that firms facing negative environmental publicity enhance sustainability communication to restore legitimacy rather than improve actual performance. Lyon and Montgomery (2015) similarly argue that extensive media attention can trigger “greenwashing under pressure,” as firms signal conformity with social norms while masking weak internal practices. Long et al. (2025) report that social media pressure significantly increases corporate hypocrisy and ESG greenwashing, as firms adopt symbolic disclosure strategies to mitigate reputational damage rather than undertake substantive sustainability reforms.

Synthesis of these studies suggests that media scrutiny conditions the relationship between SCs and ESG decoupling. Although SCs may support substantive ESG practices under low scrutiny, high media attention increases reputational and signaling pressures, encouraging firms to prioritize external ESG communication over internal improvements. Media scrutiny is expected to strengthen ESG decoupling, particularly in dimensions characterized by higher measurement ambiguity, leading to the following hypotheses:

Hypothesis 5. *Media scrutiny positively moderates the relationship between SCs and ESG decoupling, such that the positive association between SCs and ESG decoupling is stronger when media scrutiny is high.*

Hypothesis 5a. *Media scrutiny negatively moderates the relationship between SCs and environmental decoupling, such that the association is weaker when media scrutiny is high.*

Hypothesis 5b. *Media scrutiny positively moderates the relationship between SCs and social decoupling, such that the positive association is stronger when media scrutiny is high.*

Hypothesis 5c. *Media scrutiny positively moderates the relationship between SCs and governance decoupling, such that the positive association is stronger when media scrutiny is high.*

2.6.3 | Institutional Ownership

According to Bushee (1998), “institutional ownership refers to the proportion of a firm’s outstanding shares held by institutional investors such as mutual funds, pension funds, insurance companies, and hedge funds.” Institutional investors possess substantial financial and analytical resources, along with strong incentives to monitor corporate governance and sustainability performance (Gillan and Starks 2007). Consequently, institutional ownership can moderate the effectiveness of SCs in promoting substantive ESG improvements versus symbolic ESG disclosures, thereby influencing the extent of ESG decoupling.

When institutional ownership is high and long-term oriented, institutional investors act as active monitors who demand transparency, accountability, and genuine sustainability outcomes (Erhemjamts and Huang 2019). Under such conditions, SCs are more likely to promote substantive ESG actions by strengthening governance systems, integrating sustainability into strategic

decisions, and allocating resources toward meaningful environmental and social initiatives (Cepêda et al. 2025). This disciplining effect reduces the gap between ESG disclosures and internal ESG performance, mitigating overall ESG decoupling.

Conversely, when institutional ownership is transient, the moderating effect may work in the opposite direction. Short-term institutional investors often prioritize quarterly returns and stock price performance (Bushee 1998), which may create incentives for SCs to focus on externally visible ESG activities that appeal to investors and rating agencies rather than substantive internal improvements (Fu et al. 2025). In such cases, SCs may inadvertently facilitate symbolic compliance, producing ESG disclosures that maintain legitimacy without corresponding substantive change.

Two theoretical lenses help explain these dynamics. Agency theory suggests that long-term institutional investors, as powerful shareholders, can mitigate managerial opportunism by enforcing stronger monitoring and governance (Drobetz et al. 2024). SCs operating under the oversight of these investors are more likely to drive authentic ESG initiatives. In contrast, stakeholder theory highlights that managers respond to dominant stakeholder pressures to maintain legitimacy (Freeman 1984). When short-term investors apply financial rather than ethical pressure, SCs may emphasize symbolic ESG reporting, increasing the risk of decoupling.

The moderating effect of institutional ownership may also vary across ESG dimensions, consistent with the discussion in Section 2.4. Environmental performance is more likely to be disciplined by long-term institutional investors, thereby reducing environmental decoupling, whereas social and governance dimensions are less easily monitored, and so the moderating effect of institutional ownership may be weaker in these areas.

Empirical evidence supports this nuanced perspective. Song et al. (2024) find that long-term institutional investors are associated with higher-quality ESG performance, whereas short-term investors are associated with impression management and symbolic reporting. Sun and Zhao (2024) show that institutional investor heterogeneity shapes corporate ESG strategies: Dedicated long-term investors promote substantive changes, whereas transient investors amplify symbolic behavior and ESG decoupling.

Overall, institutional ownership serves as a key external governance mechanism that moderates the impact of SCs on ESG decoupling. High levels of long-term institutional ownership strengthen the ability of SCs to promote substantive ESG performance and reduce decoupling, whereas transient or short-term ownership may weaken this effect, increasing reliance on symbolic disclosures. This provides a theoretical basis for the following hypotheses:

Hypothesis 6. *Institutional ownership negatively moderates the relationship between SCs and ESG decoupling, such that SCs reduce ESG decoupling more effectively when institutional ownership is high.*

Hypothesis 6a. *Institutional ownership negatively moderates the relationship between SCs and environmental*

decoupling, such that the effect of SCs in reducing environmental decoupling is stronger with high long-term institutional ownership.

Hypothesis 6b. *Institutional ownership negatively moderates the relationship between SCs and social decoupling, such that SCs reduce social decoupling more effectively with high long-term institutional ownership.*

Hypothesis 6c. *Institutional ownership negatively moderates the relationship between SCs and governance decoupling, such that SCs reduce governance decoupling more effectively under strong long-term institutional monitoring.*

3 | Methodology

3.1 | Data and Sample Construction

The sample comprises publicly listed firms in the US capital market from 2002 to 2021. Data were collected from multiple sources to ensure comprehensive coverage of board characteristics, financial information, and ESG performance. Board committee data were obtained from BoardEx, which provides detailed information on board structure, director profiles, and committee composition. Financial and accounting variables were retrieved from Compustat by merging BoardEx data with the Compustat database using the WRDS linking table. Data on analyst coverage, institutional ownership, and media scrutiny were collected from IBES, FactSet, and RepRisk.

The initial dataset includes 90,852 firm-year observations for all firms covered in the BoardEx database. After excluding observations that could not be successfully matched with Compustat, the sample was reduced to 69,097 firm-year observations. ESG-related information was then merged from the Refinitiv ESG database, which is part of the London Stock Exchange Group and provides globally comparable firm-level ESG performance and disclosure indicators and is widely used in empirical studies (e.g., Dyck et al. 2019). Unlike MSCI ESG (KLD), the Refinitiv database distinguishes between internal ESG performance and external ESG disclosure, enabling the construction of ESG decoupling measures. Merging the BoardEx–Compustat data with Refinitiv ESG scores using CUSIP identifiers resulted in a final sample of 20,038 firm-year observations across 2759 unique firms. The full sample selection procedure is presented in Table 1.

The substantial reduction in observations during the merging process raises the possibility of sample selection bias. To address this concern, a Heckman two-stage selection model is employed as a robustness check.

3.2 | Measurement of Variables

3.2.1 | Dependent Variable

We use ESG decoupling as the main dependent variable. ESG decoupling captures the extent to which firms' external ESG disclosures (claims) exceed their internal ESG practices (policies). This measure is widely used in the literature to assess symbolic ESG

TABLE 1 | Sample selection process.

Selection process	Firm-year obs.	Number of unique firms
US listed firms covered by the BoardEx universe over 2002–2021	90,852	9542
Less: unable to be merged with Compustat to get financial data	–21,755	–1862
Less: unable to be merged with Refinitiv ESG	–49,059	–4921
Sample for main analysis	20,038	2759

communication, as it reflects situations where firms present a positive sustainability image without fully implementing corresponding internal practices (Hawn and Ioannou 2016; Boiral 2013).

The overall ESG decoupling measure (DE_ESG) is calculated as the difference between the normalized sum of external ESG actions and the normalized sum of internal ESG actions, where normalization is based on the natural logarithm of total assets to control for firm size and allow comparability across firms:

$$DE_ESG_{i,t} = \frac{\text{Sum of external actions}_{i,t}}{\ln(\text{Total assets}_{i,t})} - \frac{\text{Sum of internal actions}_{i,t}}{\ln(\text{Total assets}_{i,t})} \quad (1)$$

A positive value of DE_ESG indicates that external disclosures exceed internal practices, suggesting a higher level of symbolic ESG engagement. This difference-based specification captures both the direction and magnitude of the disclosure–performance gap, which is central to the concept of ESG decoupling.

The construction of this measure is grounded in the prior literature that distinguishes between ESG disclosure and ESG performance. For instance, Hawn and Ioannou (2016) differentiate between symbolic CSR engagement (external communication) and substantive CSR actions (internal implementation). Similarly, Marquis et al. (2016) conceptualized greenwashing as the divergence between symbolic disclosure and substantive practices. In the environmental domain, Clarkson et al. (2008) distinguish between objective environmental performance and voluntary disclosure, whereas Tariq (2025) demonstrates that firms may engage in symbolic reporting practices, increasing disclosures without corresponding improvements in internal operational practices. Building on this literature, our measure directly operationalizes ESG decoupling as the gap between external claims and internal action.

In addition to the aggregate measure, we construct dimension-specific decoupling variables for environmental (DE_E), social (DE_S), and governance (DE_G) activities using the same approach. This disaggregation is important because prior studies suggest that ESG dimensions differ in terms of measurability and susceptibility to symbolic behavior, with environmental indicators being relatively more objective, whereas social and governance dimensions are more qualitative and

prone to managerial discretion (Tsang et al. 2023). External and internal actions are identified separately for each ESG dimension.

We also examine external and internal ESG indicators (measures) separately as dependent variables to capture the absolute level of firms' ESG engagement following methodological approaches in the ESG decoupling literature that distinguish between external disclosure and internal practice measures (e.g., Cepêda et al. 2025; Loko and Schiehl 2025).

The overall external ESG indicator (*External*) represents the total number of ESG-related claims or disclosures made by a firm in a given year, normalized by the natural logarithm of total assets:

$$External_{i,t} = \frac{\text{Sum of external disclosure indicators}_{i,t}}{\ln(\text{Total assets}_{i,t})} \quad (2)$$

Similarly, the overall internal ESG indicator (*Internal*) captures the extent of firms' ESG-related internal policies and practices, also normalized by the natural logarithm of total assets:

$$Internal_{i,t} = \frac{\text{Sum of internal policy and practice indicators}_{i,t}}{\ln(\text{Total assets}_{i,t})} \quad (3)$$

This approach strengthens the robustness of our analysis by enabling us to determine whether ESG decoupling is driven by increases in external disclosures, decreases in internal practices, or a combination of both.

For the ESG dimensions, we compute dimension-specific external and internal ESG action measures using the same normalization approach as in Equations (2) and (3). External and internal actions are summed separately for each ESG dimension and scaled by the natural logarithm of total assets.

3.2.2 | Independent Variables

The primary independent variable is the presence of an SC, which represents a formal board-level mechanism responsible for overseeing a firm's ESG strategy, reporting, and performance. SCs are identified using the BoardEx database and through disclosures in annual reports and proxy statements, consistent with prior literature (e.g., Driss et al. 2024). The variable is coded as a binary indicator:

$$SC_{i,t} = \begin{cases} 1, & \text{if firm } i \text{ has a sustainability committee in year } t \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

This binary specification is theoretically grounded in corporate governance and agency theory, where the existence of a formal board committee reflects a structural shift in monitoring and oversight rather than a marginal change in intensity (e.g., Adams and Ferreira 2007; Klein 2002). In this context, the establishment of an SC signals the adoption of a dedicated governance

framework for ESG issues, which is qualitatively different from firms without such a structure.

In the governance and ESG literature, board committees are commonly operationalized as binary variables capturing their presence or absence, as the key distinction lies in whether formal oversight exists (Liao et al. 2015). Although alternative measures such as committee size, independence, or meeting frequency may capture variations in effectiveness, such data are often not consistently available across firms and time and may introduce measurement error.

From an econometric perspective, the use of a binary independent variable in a regression with a continuous dependent variable (ESG decoupling) is appropriate and widely applied. The estimated coefficient on SC reflects the average difference in ESG decoupling between firms with and without an SC, holding other factors constant.

Firms with SCs are expected to have more structured ESG reporting systems and engagement practices, which may influence both ESG decoupling and disclosure behavior.

3.2.3 | Moderating Variables

We consider three external monitoring factors—analyst coverage, institutional ownership, and media scrutiny—as moderating variables to assess whether these factors influence the relationship between SCs and ESG decoupling.

Analyst coverage (*Analyst*) is measured as the natural logarithm of one plus the number of analyst earnings forecasts at the fiscal year-end, following Yu (2008):

$$Analyst = \ln(1 + \text{Number of analyst forecasts}) \quad (5)$$

Institutional ownership (*IO%*) is measured as the proportion of total shares held by institutional investors, following Dyck et al. (2019):

$$IO\% = \frac{\text{Total institutional ownership}}{\text{Total market value of shares}} \times 100 \quad (6)$$

Media scrutiny (*MS*) of ESG-related risks is measured using the RepRisk Index (*RRI*). We construct two measures representing negative ESG-related news exposure. Each measure is scaled by firm size using the natural logarithm of total assets, following Hawn (2021):

$$RRI_Mean = \frac{\ln(1 + \text{Yearly mean RRI value})}{\ln(\text{Total assets})} \quad (7)$$

$$RRI_Max = \frac{\ln(1 + \text{Yearly maximum RRI value})}{\ln(\text{Total assets})} \quad (8)$$

3.2.4 | Control Variables

To account for firm-level factors that may influence ESG decoupling and disclosure practices, we include several control variables commonly used in the literature. These variables capture financial, operational, and governance characteristics that may affect firms' ESG behavior. Specifically, we include the following:

- Firm size (*Size*) is measured as the natural logarithm of total assets. Larger firms may face greater public scrutiny and possess more resources to engage in ESG activities.
- Cash flow (*CFO*) is calculated as cash flows from operations divided by total assets, capturing the internal liquidity available to support ESG initiatives.
- Leverage is measured as total liabilities divided by total assets. More highly leveraged firms may have fewer discretionary resources and face constraints in ESG investments.
- Profitability or return on assets (*ROA*) is measured as net income divided by total assets. More profitable firms may allocate additional resources to ESG practices.
- Sales growth is the percentage change in sales from the previous year, capturing operational expansion that may influence ESG engagement.
- Book-to-market ratio (*BTM*): the ratio of the book value of equity to the market value of equity, reflecting valuation characteristics and growth opportunities.
- Net property, plant, and equipment (*Net PPE*) is the net PPE scaled by total assets, capturing asset intensity and potential environmental impact.
- Board size is measured as the natural logarithm of the total number of directors on the board. Although most studies use the raw count, logging board size reduces skewness and mitigates the influence of outliers in the distribution

(Anderson et al. 2004; Guest 2009). Using the logged value does not affect the substantive interpretation but improves the robustness of regression estimates.

- CEO duality is an indicator variable coded as 1 if the CEO also serves as the board chair and 0 otherwise, capturing governance oversight structure.
- Board independence is the ratio of independent directors to total board members, reflecting the board's monitoring ability and potential influence on ESG practices.
- Female director is the ratio of female directors to total board members, capturing gender diversity, which may influence ESG priorities.

Appendix A provides definitions for all variables used in this study.

3.3 | Econometric Model

To examine the impact of SCs on ESG decoupling, we estimate the following baseline panel regression model:

$$DE_ESG_{i,t} = \beta_0 + \beta_1 SC_{i,t} + \beta_m Controls_{i,t} + \beta_n Fixed\ Effects_{i,t} + \epsilon_{i,t} \quad (9)$$

where $DE_ESG_{i,t}$ represents the overall ESG decoupling of firm i in year t . $SC_{i,t}$ is the main independent variable, coded as 1 if the firm has an SC in year t and 0 otherwise. $Controls_{i,t}$ is the

TABLE 2 | Summary statistics of variables in main analysis.

Variable	N	Mean	p50	p25	p75	SD
SC	20,038	0.126	0.000	0.000	0.000	0.332
ESG score	20,038	0.380	0.354	0.246	0.493	0.175
DE_ESG	20,038	-0.904	-0.875	-1.148	-0.631	0.392
DE_E	20,038	-0.075	0.000	-0.144	0.000	0.140
DE_S	20,038	-0.079	-0.093	-0.213	0.000	0.198
DE_G	20,038	-0.749	-0.708	-0.939	-0.501	0.327
Ind_Envrn	20,038	0.482	0.000	0.000	1.000	0.500
CFO	20,038	0.045	0.069	0.015	0.116	0.156
Firm size	20,038	8.174	8.181	7.015	9.384	1.820
Leverage	20,038	0.605	0.606	0.439	0.782	0.249
Sales growth	20,038	0.133	0.067	-0.014	0.175	0.409
BTM	20,038	0.458	0.371	0.194	0.639	0.395
Net PPE	20,038	0.233	0.142	0.050	0.348	0.238
Board size	20,038	2.243	2.303	2.079	2.398	0.261
CEO duality	20,038	0.171	0.000	0.000	0.000	0.376
Board independence	20,038	0.820	0.857	0.727	0.917	0.146
Female director	20,038	0.182	0.167	0.111	0.250	0.113

Note: This table presents the descriptive statistics for our main sample, including 20,038 firm-year observations (2759 firms) during our sample period, 2002–2021. All variables are defined in Appendix A.

vector of control variables. *Fixed Effects*_{*i,t*} includes year and industry fixed effects to account for unobserved heterogeneity across time and sectors. $\varepsilon_{i,t}$ is the error term.

To examine the moderating effects of external monitoring factors, such as analyst coverage, institutional ownership, and media scrutiny, we incorporate interaction terms into the model:

$$DE_ESG_{i,t} = \beta_0 + \beta_1 SC_{i,t} + \beta_2 Mod_{i,t} + \beta_3 (SC_{i,t} \times Mod_{i,t}) + \beta_m Controls_{i,t} + \beta_n Fixed\ Effects_{i,t} + \varepsilon_{i,t} \quad (10)$$

where *Mod*_{*i,t*} represents each moderating variable: analyst coverage (*Analyst*), institutional ownership (*IO%*), and media scrutiny (*RRI_Mean* or *RRI_Max*). The coefficient β_3 captures how the relationship between SCs and ESG decoupling varies with different levels of external monitoring.

We further estimate regressions using dimension-specific ESG decoupling measures (*DE_E*, *DE_S*, and *DE_G*), as well as separate external and internal ESG actions (*External* and *Internal*) as dependent variables. These additional models follow the same specification to identify which ESG dimensions and activity types are most sensitive to the presence of SCs and external scrutiny. All regressions are estimated using firm-level panel data with year and industry fixed effects to control for unobserved heterogeneity. Robust standard errors clustered at the firm level are used to account for potential serial correlation and heteroskedasticity.

4 | Results and Discussion

4.1 | Descriptive Statistics

Table 2 presents the descriptive statistics for the key variables used in the main analysis, based on 20,038 firm-year observations from 2759 US publicly listed firms between 2002 and 2021. The mean value of SC is 0.126, with a standard deviation (SD) of 0.332, indicating that approximately 13% of firms have established a dedicated SC. This relatively low proportion suggests that the formal integration of sustainability oversight into board structures remains limited but is gradually increasing.

The average *ESG Score* is 0.380 (SD = 0.175), reflecting moderate ESG performance across firms. The mean value of ESG decoupling (*DE_ESG*) is −0.904 (SD = 0.392). The negative mean indicates that, on average, firms' internal ESG actions slightly exceed their external disclosures, suggesting that firms may focus more on implementing ESG practices than on reporting them.

At the pillar level, environmental decoupling (*DE_E*) and social decoupling (*DE_S*) have small negative means (−0.075 and −0.079, respectively), whereas governance decoupling (*DE_G*) is substantially more negative (−0.749). These negative averages indicate that internal governance practices generally exceed reported ESG disclosures, particularly in the governance dimension.

Overall, the descriptive statistics reveal substantial heterogeneity across firms in their sustainability governance structures and ESG performance. The negative averages for the decoupling

measures highlight that, on average, firms tend to have stronger internal ESG actions than external reporting. These unconditional averages provide an overview of the sample, whereas our regression analysis further examines how SCs influence ESG decoupling after controlling for firm- and industry-level characteristics.

4.2 | Trend Analysis of SCs

To examine the adoption of SCs over time, we first present descriptive statistics on the yearly distribution of firms with SCs in our sample (see Table 3). The results show that SC adoption fluctuated over the study period. In the early 2000s, only a small proportion of firms had SCs, ranging from 14% to 17%. Adoption declined during 2015–2017, reaching a low of 8.4% in 2017. However, a noticeable increase occurred after 2018, with the proportion of firms with SCs rising to 15.65% in 2021. Overall, 2521 firm-year observations out of 20,038 (12.58%) included SCs, representing 416 unique firms (15.08% of the total sample).

TABLE 3 | Yearly distribution of firms with sustainability committee.

	Firms with SC	All firms	
Year	N	N	Percentage
2002	40	232	17.24%
2003	40	239	16.74%
2004	46	327	14.07%
2005	59	376	15.69%
2006	63	378	16.67%
2007	67	402	16.67%
2008	79	515	15.34%
2009	86	573	15.01%
2010	96	615	15.61%
2011	104	630	16.51%
2012	108	634	17.03%
2013	110	648	16.98%
2014	115	665	17.29%
2015	142	1079	13.16%
2016	149	1547	9.63%
2017	168	2000	8.40%
2018	185	2173	8.51%
2019	226	2348	9.63%
2020	302	2510	12.03%
2021	336	2147	15.65%
Total	2521	20,038	12.58%
Unique firms	416	2759	15.08%

TABLE 4 | Industrial sector distribution of firms with sustainability committee.

Sector (FTSE International classification)	Firms with SC	All firms	Percentage
	N	N	
Aerospace and defense	73	209	34.93%
Automobiles and parts	80	422	18.96%
Banks	73	1697	4.30%
Beverages	26	125	20.80%
Business services	45	784	5.74%
Chemicals	243	551	44.10%
Clothing and personal products	58	360	16.11%
Construction and building material	45	632	7.12%
Consumer services	3	122	2.46%
Containers and packaging	16	187	8.56%
Diversified industrials	30	151	19.87%
Education	0	73	0.00%
Electricity	39	122	31.97%
Electronic and electrical equipment	100	1384	7.23%
Engineering and machinery	129	937	13.77%
Food and drug retailers	22	128	17.19%
Food producers and processors	114	518	22.01%
Forestry and paper	74	135	54.81%
General retailers	80	793	10.09%
Health	13	1000	1.30%
Household products	86	314	27.39%
Information technology hardware	22	517	4.26%
Insurance	33	474	6.96%
Investment companies	0	9	0.00%
Leisure and hotels	128	780	16.41%
Leisure goods	38	139	27.34%
Life assurance	0	16	0.00%

(Continues)

TABLE 4 | (Continued)

Sector (FTSE International classification)	Firms with SC	All firms	Percentage
	N	N	
Media and entertainment	0	288	0.00%
Mining	109	216	50.46%
Oil and gas	274	950	28.84%
Pharmaceuticals and biotechnology	67	1468	4.56%
Private equity	0	17	0.00%
Publishing	17	130	13.08%
Real estate	4	306	1.31%
Renewable energy	4	145	2.76%
Software and computer services	18	1156	1.56%
Specialty and other finance	39	761	5.12%
Steel and other metals	46	219	21.00%
Telecommunication services	42	416	10.10%
Tobacco	21	50	42.00%
Transport	13	570	2.28%
Utilities—other	297	767	38.72%
Total	2521	20,038	12.58%

We analyzed SC adoption across industrial sectors using the FTSE International classification (see Table 4). The prevalence of SCs varies across industries. Sectors such as forestry and paper (54.81%), mining (50.46%), chemicals (44.10%), and tobacco (42.00%) show the highest adoption rates, indicating a stronger focus on sustainability in resource-intensive industries. In contrast, sectors including education, life assurance, investment companies, and media and entertainment show zero adoption, whereas other service-oriented sectors report very low percentages, reflecting differences in regulatory pressures, stakeholder expectations, and sustainability priorities across industries.

These descriptive results highlight two key trends: the overall adoption of SCs is gradually increasing, especially after 2018, and adoption is highly sector-specific, with resource-intensive and environmentally sensitive industries more likely to establish SCs.

4.3 | Correlation Analysis

Table 5 presents the pairwise correlation coefficients for the key variables used in the main analysis. The correlations are reported

TABLE 5 | Pairwise correlation matrix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) SC	1.000																
(2) ESG score	0.2717*	1.000															
(3) DE_ESG	0.1678*	0.2363*	1.000														
(4) DE_E	-0.1834*	-0.4156*	0.1345*	1.000													
(5) DE_S	0.0741*	0.0895*	0.5603*	-0.0339*	1.000												
(6) DE_G	0.2353*	0.4083*	0.7931*	-0.2487*	0.0900*	1.000											
(7) Ind_Envrm	0.0945*	0.0510*	-0.1409*	-0.1706*	0.0313*	-0.1136*	1.000										
(8) CFO	0.0777*	0.1952*	0.2829*	-0.1478*	-0.017	0.4048*	-0.0792*	1.000									
(9) Firm size	0.2594*	0.4529*	0.6794*	-0.1853*	0.1350*	0.6106*	-0.1730*	0.3442*	1.000								
(10) Leverage	0.0776*	0.1105*	0.1939*	-0.009	0.0309*	0.2186*	-0.2806*	-0.0606*	0.3183*	1.000							
(11) Sales growth	-0.0694*	-0.1159*	-0.0764*	0.0926*	0.0330*	-0.1533*	0.0433*	-0.1238*	-0.1534*	-0.0986*	1.000						
(12) BTM	0.007	-0.0782*	0.0627*	0.0441*	-0.024	0.0690*	-0.1064*	-0.0342*	0.1314*	-0.0901*	-0.1096*	1.000					
(13) Net PPE	0.2160*	0.0791*	0.0775*	-0.1273*	-0.0345*	0.1685*	-0.002	0.1616*	0.1193*	-0.008	-0.0839*	0.0916*	1.000				
(14) Board size	0.2204*	0.3644*	0.4244*	-0.1434*	0.0796*	0.5209*	-0.0980*	0.1701*	0.6369*	0.2828*	-0.1178*	0.0361*	0.0385*	1.000			
(15) CEO duality	0.0867*	0.0701*	0.1815*	-0.016	0.0300*	0.2049*	-0.0293*	0.0833*	0.2438*	0.1229*	-0.0617*	0.002	0.0405*	0.2034*	1.000		
(16) Board independence	0.1713*	0.3425*	-0.021	-0.1502*	0.002	0.0393*	0.1040*	0.003	0.1520*	0.0779*	-0.0712*	-0.025	0.0309*	0.1300*	0.0565*	1.000	
(17) Female director	0.1383*	0.3811*	-0.0321*	-0.1797*	-0.1516*	0.1296*	-0.0646*	0.0307*	0.1472*	0.1033*	-0.0486*	-0.0712*	-0.014	0.1685*	-0.003	0.2606*	1.000

Note: This table reports the pairwise coefficient estimates of variables for main analysis. All continuous variables are winsorized at the 1% and 99% levels. Variable definitions are provided in Appendix A.
* $p < 0.01$.

for descriptive purposes to illustrate the strength and direction of bivariate relationships. Statistical inferences regarding hypotheses are evaluated in the subsequent regression analyses.

4.4 | Multicollinearity Test

To ensure the reliability of our regression estimates, we conducted a variance inflation factor (VIF) test to examine potential multicollinearity among the independent and control variables.

Table 6 reports the VIF for all explanatory variables included in the regressions. The results indicate that all VIF values are below 3, with a mean VIF of 1.51. The main independent variable, *SC*, has a VIF of 1.20. Because all values are well below the commonly accepted threshold of 10, multicollinearity is not a concern in this dataset.

4.5 | Baseline Regression Results

To examine the effect of SCs on ESG decoupling, we estimate an ordinary least squares (OLS) regression model. The results presented in Table 7 show that SCs are positively and significantly associated with overall ESG decoupling (DE_ESG ; $\beta = 0.0668$, $p < 0.01$). This result supports *Hypothesis 1*: SCs have a significant positive influence on overall ESG decoupling. This finding indicates that firms with SCs tend to exhibit a higher degree of misalignment between reported and actual ESG performance, consistent with prior studies that argue governance structures may be used to signal commitment without corresponding operational changes (Kahan and Rock 2014).

From a theoretical standpoint, this finding aligns with legitimacy theory and symbolic management theory, which propose that firms often adopt formal governance structures to gain external legitimacy and satisfy stakeholder expectations rather than to achieve genuine performance improvements. In this context, the establishment of SCs can be viewed as a strategic

response to institutional pressures from investors, regulators, and rating agencies demanding greater ESG accountability. SCs may therefore function as symbolic devices that signal ethical or responsible conduct rather than as mechanisms that promote substantive ESG progress.

Empirical research also supports these theoretical assertions. Rodrigue et al. (2013) and Boiral et al. (2019) document that many firms engage in symbolic sustainability practices to maintain legitimacy and enhance reputation without fundamentally changing operational behaviors. Baraibar-Diez and Odriozola (2019) similarly report that CSR committees often improve the quality of ESG reporting but do not necessarily enhance actual ESG performance. These findings suggest that the positive relationship between SCs and overall ESG decoupling reflects a legitimacy-driven motive, in which SCs attract ESG-conscious investors and mitigate reputational risk even when substantive progress is limited.

We next examine the subdimensions of ESG decoupling. The results indicate that SCs have a negative effect on environmental decoupling (DE_E ; $\beta = -0.0097$, $p < 0.10$). This result supports *Hypothesis 1a*: SCs have a significant negative influence on environmental decoupling. One explanation is that environmental performance is more quantifiable, more heavily regulated, and more visible to external stakeholders. Metrics such as emissions, energy use, and waste generation are subject to standardized measurement and regulatory audits, making symbolic reporting riskier and more difficult to maintain (He 2025). As a result, SCs have strong incentives to ensure genuine environmental improvements, supported by enhanced monitoring, clearer environmental targets, and more effective implementation oversight. Environmental initiatives also often yield direct economic benefits, such as cost savings, operational efficiencies, and reduced regulatory exposure, further motivating committees to pursue substantive practices (Driss et al. 2024). These arguments are consistent with evidence that firms with dedicated governance structures improve environmental transparency and compliance (Rodrigue et al. 2013). Li et al. (2023) demonstrate that the presence and effectiveness of SCs are significantly associated with higher environmental performance, reinforcing the conclusion that SCs help reduce environmental decoupling.

In contrast, SCs are positively associated with social decoupling (DE_S ; $\beta = 0.0527$, $p < 0.01$) and governance decoupling (DE_G ; $\beta = 0.0238$, $p < 0.01$). These results support *Hypotheses 1b* and *1c*: SCs have a significant positive influence on social and governance decoupling, respectively. Social and governance performance is more qualitative, less standardized, and less rigorously regulated, making these dimensions more vulnerable to symbolic actions. Social indicators such as community engagement, employee well-being, and diversity often rely on narrative disclosures or firm-reported practices that are difficult for external stakeholders to verify (Eccles et al. 2020). Governance indicators, including ethics training, internal controls, and board independence, can also be influenced by superficial or symbolic adjustments rather than structural reforms (Westphal and Park 2020). Because these areas lack strict measurement frameworks and are easier to manipulate, firms may strategically use SCs as legitimacy-seeking tools to enhance reporting while avoiding costly reforms, heightening social and governance decoupling.

TABLE 6 | Variance inflation factor (VIF) diagnostics test.

Variable	VIF	1/VIF
SC	1.20	0.833
CFO	1.50	0.667
Firm size	2.50	0.400
Leverage	1.40	0.714
Sales growth	1.20	0.833
BTM	1.10	0.909
Net PPE	1.20	0.833
Board size	1.80	0.556
CEO duality	1.10	0.909
Board independence	1.20	0.909
Female director	1.30	0.769
Mean VIF	1.51	

TABLE 7 | Baseline regression results.

	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.0668*** (6.30)	−0.00972* (−1.71)	0.0527*** (7.65)	0.0238*** (4.30)
CFO	−0.158** (−2.12)	−0.00409 (−0.13)	−0.100** (−2.19)	−0.0563 (−1.03)
Firm size	0.101*** (20.20)	−0.000811 (−0.36)	−0.00976*** (−2.92)	0.111*** (36.89)
Leverage	−0.0109 (−0.68)	0.0115* (1.69)	−0.00454 (−0.43)	−0.00946 (−0.95)
Sales growth	0.0154*** (3.46)	0.00174 (1.11)	0.0147*** (5.00)	−0.00250 (−0.85)
BTM	0.00165 (0.22)	0.00909*** (2.63)	−0.00789 (−1.62)	0.000874 (0.21)
Net PPE	−0.0264 (−0.76)	0.0172 (1.20)	−0.0483** (−2.17)	0.00694 (0.32)
Board size	−0.0447*** (−3.09)	−0.0118* (−1.77)	−0.0213** (−2.23)	−0.00985 (−1.21)
CEO duality	−0.0103 (−1.44)	−0.00255 (−0.71)	−0.00168 (−0.37)	−0.00625* (−1.69)
Board independence	−0.111*** (−6.34)	−0.00810 (−0.99)	−0.0120 (−1.01)	−0.0899*** (−8.82)
Female director	−0.428*** (−15.73)	−0.0343*** (−2.68)	−0.448*** (−23.87)	0.0562*** (3.56)
<i>N</i>	19,896	19,896	19,896	19,896
Adj. <i>R</i> -squared	0.747	0.560	0.545	0.884
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

p* < 0.1.*p* < 0.05.****p* < 0.01.

Overall, the findings support *Hypothesis 1* and its sub-hypotheses (*Hypotheses 1a–1c*), showing that SCs are positively associated with overall ESG decoupling while simultaneously promoting substantive environmental alignment (*Hypothesis 1a*) and facilitating symbolic practices in social and governance reporting (*Hypotheses 1b* and *1c*).

4.6 | Effect of SCs on External and Internal ESG Action

Next, we examine the effect of SCs on external and internal ESG actions, including overall and dimension-specific (E, S, G) activities. The results in Table 8 show that SCs are

positively and significantly associated with overall external ESG actions (*EX*; $\beta = 0.0692$, $p < 0.01$), providing support for *Hypothesis 2*: SCs have a significant positive influence on external ESG actions. This finding suggests that firms with SCs tend to engage more actively in externally visible ESG initiatives aimed at enhancing legitimacy with key stakeholders. External ESG actions, such as public CSR campaigns, community programs, and sustainability reports, are more observable and easier to communicate. As a result, SCs may strategically prioritize these initiatives as cost-effective tools for signaling ESG commitment. In contrast, SCs have a smaller and statistically insignificant effect on overall internal ESG actions (*IN*; $\beta = 0.00865$), indicating that there is no clear evidence that SCs influence internal ESG actions, which is consistent with

TABLE 8 | Effect of sustainability committee on external and internal ESG actions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	External	Internal	EX_E	IN_E	EX_S	IN_S	EX_G	IN_G
SC	0.0692*** (5.78)	0.00865 (0.74)	0.0121** (2.56)	0.0227*** (3.64)	0.0319*** (4.82)	-0.0192*** (-3.03)	0.0254*** (6.58)	0.00340 (0.84)
CFO	-0.187** (-2.33)	-0.0356 (-0.42)	-0.0410 (-1.35)	-0.0402 (-1.11)	-0.110*** (-2.62)	0.00180 (0.04)	-0.0267 (-0.90)	0.0264 (0.57)
Firm size	0.000118 (0.02)	-0.102*** (-18.69)	0.0106*** (5.22)	0.0112*** (4.09)	-0.0165*** (-5.82)	-0.00572* (-1.92)	0.00614*** (3.46)	-0.104*** (-42.64)
Leverage	-0.0135 (-0.78)	-0.00529 (-0.29)	0.000675 (0.10)	-0.0109 (-1.27)	-0.000640 (-0.07)	0.00368 (0.38)	-0.0191*** (-3.47)	-0.00894 (-1.11)
Sales growth	-0.00674 (-1.53)	-0.0219*** (-4.39)	-0.00483*** (-3.10)	-0.00641*** (-3.05)	0.000872 (0.37)	-0.0147*** (-5.14)	-0.00320** (-2.36)	-0.000842 (-0.33)
BTM	-0.0256*** (-3.30)	-0.0292*** (-3.72)	-0.00477 (-1.59)	-0.0134*** (-3.25)	-0.0113*** (-2.76)	-0.00347 (-0.82)	-0.0103*** (-4.12)	-0.0111*** (-3.33)
Net PPE	0.0582* (1.69)	0.0787** (2.12)	0.0208 (1.57)	-0.000114 (-0.01)	0.0266 (1.45)	0.0687*** (3.39)	0.0119 (1.01)	0.00409 (0.23)
Board size	0.0686*** (4.50)	0.109*** (7.03)	0.0164*** (2.79)	0.0282*** (3.53)	0.0269*** (3.28)	0.0471*** (5.57)	0.0220*** (4.23)	0.0305*** (4.76)
CEO duality	-0.00146 (-0.19)	0.00905 (1.20)	-0.00614** (-2.03)	-0.00380 (-0.95)	0.00337 (0.81)	0.00416 (1.03)	0.00159 (0.61)	0.00736*** (2.74)
Board independence	0.0455** (2.49)	0.153*** (7.93)	0.00797 (1.13)	0.0152 (1.54)	0.0258*** (2.60)	0.0382*** (3.56)	0.0103 (1.62)	0.0989*** (12.18)
Female director	0.221*** (7.76)	0.661*** (22.41)	0.0879*** (7.78)	0.122*** (8.17)	0.0886*** (5.73)	0.541*** (32.26)	0.0466*** (4.77)	-0.00622 (-0.50)
N	19,896	19,896	19,896	19,896	19,896	19,896	19,896	19,896
Adj. R-squared	0.764	0.801	0.694	0.741	0.736	0.731	0.682	0.905
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$.** $p < 0.05$.*** $p < 0.01$.

Hypothesis 3 that predicts a weak or negative effect. Internal ESG activities typically require substantial resource commitments, structural changes, and long-term investments in organizational practices. These activities offer fewer immediate reputational benefits, which may explain why firms with SCs focus less on internal operational improvements. These findings are consistent with prior evidence that sustainability governance structures often improve the quantity of ESG disclosure rather than the underlying quality of ESG performance (Baraibar-Diez and Odriozola 2019; Boiral et al. 2019). Overall, the results support our earlier evidence that SCs contribute to ESG decoupling by strengthening external ESG profiles without promoting comparable internal improvements, reinforcing their role as legitimacy-seeking mechanisms rather than drivers of substantive organizational change.

The results also indicate that SCs are positively associated with both external environmental actions (EX_E ; $\beta = 0.0121$, $p < 0.05$) and internal environmental actions (IN_E ; $\beta = 0.0227$, $p < 0.01$), supporting *Hypotheses 2a* and *3a*. These findings suggest that SCs play a meaningful role in advancing environmental performance. Environmental initiatives often deliver tangible benefits, including cost savings from energy efficiency, reduced regulatory risk, and enhanced reputation among stakeholders, which encourage SCs to implement and monitor environmental policies both internally and externally. Consequently, SCs appear to facilitate visible external environmental efforts and substantive internal operational improvements. This pattern aligns with our earlier finding that SCs reduce environmental decoupling and supports evidence that SCs can effectively drive improvements in environmental outcomes (Rodrigue et al. 2013; Li et al. 2023).

For social actions, SCs have a positive and significant impact on external social initiatives (EX_S ; $\beta = 0.0319$, $p < 0.01$) but a negative effect on internal social practices (IN_S ; $\beta = -0.0192$, $p < 0.01$), supporting *Hypotheses 2b* and *3b*. These results suggest that SCs place greater emphasis on outward-facing social initiatives while paying less attention to internal social reforms. External social actions, such as philanthropy, community engagement, and public social disclosures, offer high visibility and immediate legitimacy benefits. In contrast, internal social improvements, such as employee well-being programs, workforce development, and improved labor conditions, require greater resource commitments and generate more gradual returns. As a result, SCs may enhance symbolic external social efforts while neglecting deeper internal changes. This divergence reinforces our earlier finding that SCs increase social decoupling by boosting external social profiles without promoting corresponding internal improvements.

A similar pattern emerges for governance actions. SCs are positively and significantly associated with external governance actions (EX_G ; $\beta = 0.0254$, $p < 0.01$), but the effect on internal governance actions is statistically insignificant (IN_G ; $\beta = 0.00340$), supporting *Hypotheses 2c* and *3c*. These results suggest that SCs place more emphasis on visible governance practices meant to signal good governance to external audiences rather than on substantive internal governance reforms. Thus, the stronger emphasis on external governance initiatives reflects a strategic approach to reputation management, which is

consistent with the finding that SCs contribute to symbolic ESG practices in the governance dimension.

Overall, the results indicate that SCs reinforce symbolic ESG practices in the social and governance dimensions but drive substantive improvements in the environmental dimension, highlighting their complex role in shaping external and internal ESG actions.

4.7 | Robustness Checks

To ensure the reliability of our main findings, we conducted several robustness checks using alternative dependent variable measures, propensity score matching (PSM), and the Heckman two-stage method to account for potential sample selection bias and endogeneity.

4.7.1 | Alternative ESG Decoupling Measures

First, we addressed potential scaling issues in the ESG indexes. Following Hussain et al. (2018), the original external and internal ESG indexes are normalized between 0 and 1. In our sample, a small number of observations fell outside this range, so we applied two alternative calculation methods consistent with prior research (e.g., Hussain et al. 2018). The first method replaced the original scaling factor (the natural logarithm of total assets) with the total number of ESG items (21 external actions and 20 internal actions):

$$DE_ESG_{i,t}^{a1} = \frac{\text{Sum of external actions}_{i,t}}{\text{Number of items}_{i,t}^E} - \frac{\text{Sum of internal actions}_{i,t}}{\text{Number of items}_{i,t}^I} \quad (11)$$

The second method applied min-max normalization, adjusting each sum of actions relative to the annual minimum and maximum values:

$$\text{External Index}_{i,t}^{a2} = \frac{\text{Sum of external actions}_{i,t} - \text{Minimum}_t^E}{\text{Maximum}_t^E - \text{Minimum}_t^E} \quad (12)$$

$$\text{Internal Index}_{i,t}^{a2} = \frac{\text{Sum of internal actions}_{i,t} - \text{Minimum}_t^I}{\text{Maximum}_t^I - \text{Minimum}_t^I} \quad (13)$$

$$DE_ESG_{i,t}^{a2} = \text{External Index}_{i,t}^{a2} - \text{Internal Index}_{i,t}^{a2} \quad (14)$$

Using these alternative approaches, we also constructed dimension-specific decoupling measures for environmental (DE_E), social (DE_S), and governance (DE_G) activities, with sums of external and internal actions identified separately for each dimension.

The results using these alternative measures, reported in Tables 9 and 10, remain consistent with the main analysis. SCs are positively associated with overall ESG decoupling, as well as with social and governance decoupling, while maintaining a negative association with environmental decoupling.

TABLE 9 | Robustness test using alternative dependent variable measures (1).

	(1) DE_ESG ^{a1}	(2) DE_E ^{a1}	(3) DE_S ^{a1}	(4) DE_G ^{a1}
SC	0.0287*** (6.23)	−0.0220*** (−2.74)	0.0500*** (5.94)	0.0675*** (7.25)
CFO	−0.0734*** (−2.76)	−0.00908 (−0.22)	−0.102** (−2.02)	−0.0977 (−1.42)
Firm size	−0.00889*** (−4.47)	−0.0151*** (−4.67)	−0.0364*** (−9.54)	0.0216*** (4.96)
Leverage	0.000259 (0.04)	0.0229** (2.43)	0.000214 (0.02)	−0.0363*** (−2.65)
Sales growth	0.00509*** (3.55)	0.00355* (1.66)	0.0156*** (5.29)	−0.00487 (−1.41)
BTM	−0.000581 (−0.20)	0.0148*** (3.09)	−0.00295 (−0.54)	−0.0159** (−2.57)
Net PPE	−0.00956 (−0.74)	0.0284 (1.40)	−0.0823*** (−3.26)	0.00931 (0.32)
Board size	−0.0144** (−2.49)	−0.0150 (−1.57)	−0.0400*** (−3.60)	0.0146 (1.15)
CEO duality	−0.00259 (−0.84)	−0.00376 (−0.71)	−0.00361 (−0.64)	0.00110 (0.17)
Board independence	−0.0457*** (−6.71)	−0.0112 (−0.97)	−0.0302** (−2.25)	−0.0832*** (−5.45)
Female director	−0.180*** (−16.96)s	−0.0581*** (−3.28)	−0.619*** (−29.16)	0.0496** (2.10)
<i>N</i>	19,896	19,896	19,896	19,896
Adj. <i>R</i> -squared	0.524	0.610	0.559	0.672
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for robustness test using alternative dependent variable measures (replace original scaler with the number of total items).

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

4.7.2 | PSM

Next, to address concerns that differences between firms with and without SCs may bias the results, we conducted PSM. We first excluded firms that maintained SCs throughout the entire sample period, then performed nearest-neighbor matching to construct more comparable treatment and control groups.

Tables 11 and 12 present the results from the 1-to-3 and 1-to-4 matched samples, respectively. Consistent with the baseline findings, SCs remain significantly and positively associated with overall ESG decoupling, as well as with social and governance decoupling. The negative association with environmental decoupling

also persists across both matched samples. These results indicate that the observed effects are not driven by systematic differences in firm characteristics, reinforcing the robustness of our conclusions.

4.7.3 | Heckman Two-Stage Procedure and Exclusion Restriction

Finally, we addressed potential sample selection bias using the Heckman (1979) two-stage procedure. In the first-stage selection equation, we model the likelihood that a firm establishes an SC using firm-level characteristics (e.g., board size, board independence, CEO duality, and female directors) and an exclusion

TABLE 10 | Robustness test using alternative dependent variable measures (2).

	(1) DE_ESG ^{a2}	(2) DE_E ^{a2}	(3) DE_S ^{a2}	(4) DE_G ^{a2}
SC	0.0189*** (3.66)	−0.0253*** (−3.07)	0.0466*** (5.61)	0.0638*** (6.21)
CFO	−0.0780** (−2.53)	−0.0133 (−0.31)	−0.106** (−2.12)	−0.106 (−1.38)
Firm size	−0.00536** (−2.41)	−0.0103*** (−3.08)	−0.0356*** (−9.39)	0.0217*** (4.58)
Leverage	0.00189 (0.29)	0.0256*** (2.63)	−0.00350 (−0.31)	−0.0313** (−2.06)
Sales growth	0.00312* (1.94)	0.00198 (0.90)	0.0155*** (5.27)	−0.00516 (−1.32)
BTM	0.00425 (1.28)	0.0171*** (3.47)	−0.00310 (−0.57)	−0.0126* (−1.85)
Net PPE	−0.00467 (−0.32)	0.0313 (1.45)	−0.0692*** (−2.76)	0.00176 (0.05)
Board size	−0.0160** (−2.44)	−0.0126 (−1.28)	−0.0372*** (−3.39)	0.00615 (0.44)
CEO duality	−0.00146 (−0.42)	−0.00422 (−0.76)	−0.00287 (−0.52)	−0.000935 (−0.13)
Board independence	−0.0502*** (−6.50)	−0.00767 (−0.64)	−0.0285** (−2.14)	−0.119*** (−7.04)
Female director	−0.199*** (−16.80)	−0.0436** (−2.38)	−0.617*** (−29.35)	0.0455* (1.75)
<i>N</i>	19,896	19,896	19,896	19,896
Adj. <i>R</i> -squared	0.494	0.580	0.572	0.603
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for robustness test using alternative dependent variable measures (using min–max feature scaling method).

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

restriction: the yearly industry-level average ESG score. This variable influences a firm's decision to adopt an SC because firms in industries with stronger ESG norms face greater pressure to establish SCs. Importantly, it is unlikely to directly affect a specific firm's ESG decoupling, thereby satisfying the exclusion restriction requirement and supporting model identification.

In the second stage, we include the inverse Mills ratio from the first stage in the main regression to correct for selection bias. As shown in Table 13, the Heckman-adjusted coefficients remain consistent with our main results in both sign and statistical significance, confirming that sample selection bias and non-random SC adoption do not drive our findings.

Overall, these robustness checks confirm that our main findings are reliable and not sensitive to alternative measures, sample design, or potential endogeneity.

4.8 | Moderating Analysis

To further understand how SCs influence ESG decoupling, we examine whether this relationship depends on the level of external monitoring and scrutiny. Grounded in contingency theory, which emphasizes the role of contextual factors in shaping organizational practices (Donaldson 2001), and stakeholder theory, which highlights firms' strategic responses to the demands

TABLE 11 | Robustness test using PSM sample (1–3 matching).

	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.0231* (1.72)	−0.0205*** (−2.92)	0.0268*** (3.00)	0.0163** (2.28)
CFO	−0.415*** (−2.77)	0.0321 (0.49)	−0.317*** (−3.22)	−0.138 (−1.63)
Firm size	0.0833*** (9.28)	−0.00578 (−1.34)	0.000525 (0.09)	0.0879*** (17.45)
Leverage	0.0135 (0.44)	0.00844 (0.55)	−0.00853 (−0.42)	0.0147 (0.81)
Sales growth	0.00893 (0.81)	0.0117** (2.20)	0.00689 (0.96)	−0.00939 (−1.33)
BTM	−0.00668 (−0.47)	0.0154** (2.06)	−0.0208** (−2.32)	−0.000979 (−0.13)
Net PPE	0.0345 (0.63)	−0.00707 (−0.26)	−0.0439 (−1.16)	0.0824*** (2.68)
Board size	−0.0346 (−1.36)	−0.0155 (−1.22)	−0.0146 (−0.87)	−0.00767 (−0.56)
CEO duality	−0.0130 (−1.17)	−0.00367 (−0.62)	−0.00143 (−0.20)	−0.00772 (−1.29)
Board independence	−0.114*** (−3.57)	−0.00261 (−0.16)	−0.0102 (−0.48)	−0.0951*** (−5.35)
Female director	−0.437*** (−9.00)	−0.0695*** (−2.81)	−0.394*** (−11.95)	0.0288 (1.09)
<i>N</i>	6704	6704	6704	6704
Adj. <i>R</i> -squared	0.685	0.529	0.498	0.827
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for robustness test using 1–3 PSM sample.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

of multiple stakeholders (Freeman 1984), we argue that the effectiveness of SCs in mitigating or exacerbating ESG decoupling may vary under different levels of external oversight. Accordingly, we investigate whether analyst coverage, institutional ownership, and media scrutiny moderate the effect of SCs on ESG decoupling. The results are presented in Tables 14–16.

4.8.1 | Moderating Effect of Analyst Coverage

The results in Table 14 show that analyst coverage significantly strengthens the relationship between SCs and ESG decoupling. For overall ESG decoupling, the interaction term $SC \times Analyst$ is

positive and statistically significant ($\beta = 0.0365$, $p < 0.01$), providing support for *Hypothesis 4*: Analyst coverage positively moderates the relationship between SCs and ESG decoupling, leading firms with SCs to display higher levels of ESG decoupling under greater analyst scrutiny, reflecting a tendency toward impression management rather than substantive sustainability integration.

Dimension-specific results support this interpretation. The interaction term is positive and significant for social decoupling (DE_S ; $\beta = 0.0298$, $p < 0.01$) and governance decoupling (DE_G ; $\beta = 0.00873$, $p < 0.05$), providing support for *Hypotheses 4b* and *4c*. This suggests that under high analyst attention, firms with SCs are more likely to engage in symbolic social and governance

TABLE 12 | Robustness test using PSM sample (1–4 matching).

	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.0279** (2.10)	−0.0192*** (−2.76)	0.0311*** (3.52)	0.0159** (2.25)
CFO	−0.395*** (−2.75)	0.0164 (0.26)	−0.292*** (−3.12)	−0.125 (−1.55)
Firm size	0.0864*** (10.44)	−0.00440 (−1.13)	−0.000311 (−0.06)	0.0903*** (19.19)
Leverage	0.0203 (0.70)	0.00890 (0.63)	0.0000272 (0.00)	0.0123 (0.73)
Sales growth	0.00659 (0.68)	0.00924* (1.95)	0.00715 (1.08)	−0.00912 (−1.46)
BTM	−0.00505 (−0.38)	0.0139** (2.03)	−0.0127 (−1.48)	−0.00593 (−0.83)
Net PPE	0.0131 (0.25)	−0.0113 (−0.44)	−0.0575 (−1.58)	0.0780*** (2.63)
Board size	−0.0225 (−0.95)	−0.0135 (−1.15)	−0.0118 (−0.75)	0.000188 (0.01)
CEO duality	−0.00780 (−0.74)	−0.00182 (−0.33)	0.00253 (0.38)	−0.00845 (−1.49)
Board independence	−0.129*** (−4.33)	−0.000493 (−0.03)	−0.0260 (−1.31)	−0.0963*** (−5.80)
Female director	−0.430*** (−9.39)	−0.0632*** (−2.71)	−0.388*** (−12.44)	0.0262 (1.04)
<i>N</i>	7659	7659	7659	7659
Adj. <i>R</i> -squared	0.685	0.533	0.498	0.830
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for robustness test using 1–4 PSM sample.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

reporting, strengthening the positive association between SCs and decoupling in these dimensions.

In contrast, the interaction term is statistically insignificant for environmental decoupling (DE_E ; $\beta = -0.00193$, $p > 0.05$), indicating that there is no moderating effect of analyst coverage on the relationship between SCs and environmental decoupling, and thus *Hypothesis 4a* is not supported. This implies that environmental outcomes, which are more measurable and regulated, are less susceptible to analyst-driven impression management and instead require substantive actions that cannot be easily influenced by external scrutiny.

4.8.2 | Moderating Effect of Media Scrutiny

The results reported in Table 15 show that ESG-related media coverage does not moderate the relationship between SCs and overall ESG decoupling. The interaction term $SC \times RRI$ for DE_ESG is positive but statistically insignificant (Panel A: $\beta = 0.0571$; Panel B: $\beta = 0.0422$), indicating that there is no moderating effect of media scrutiny on the relationship between SCs and overall ESG decoupling, and thus *Hypothesis 5* is not supported. At the aggregate ESG level, media coverage does not appear to alter how SCs influence the gap between ESG disclosures and actual practices.

TABLE 13 | Robustness test for sample selection bias: Heckman two-stage method.

	Selection	Main model			
	SC	DE_ESG	DE_E	DE_S	DE_G
ESG score	0.812*** (8.80)				
Ind_Mean.ESG	1.108*** (4.89)				
SC		0.0635*** (6.03)	−0.0138** (−2.51)	0.0524*** (7.60)	0.0248*** (4.50)
Inverse Mills ratio		0.231*** (9.93)	0.282*** (25.31)	0.0232 (1.45)	−0.0710*** (−5.54)
Ind_Envrn	0.547*** (18.24)				
CFO	0.331 (0.73)	−0.0905 (−1.21)	0.0781** (2.56)	−0.0936** (−2.03)	−0.0770 (−1.41)
Firm size	0.145*** (13.05)	0.133*** (22.38)	0.0380*** (13.92)	−0.00658 (−1.61)	0.101*** (29.17)
Leverage	0.371*** (5.90)	0.0628*** (3.59)	0.101*** (13.36)	0.00285 (0.24)	−0.0321*** (−2.97)
Sales growth	−0.134*** (−2.64)	−0.0124** (−2.37)	−0.0322*** (−15.67)	0.0119*** (3.39)	0.00603* (1.82)
BTM	0.0463 (1.36)	0.00701 (0.93)	0.0156*** (4.60)	−0.00735 (−1.50)	−0.000774 (−0.18)
Net PPE	1.331*** (22.44)	0.222*** (5.23)	0.320*** (17.18)	−0.0234 (−0.82)	−0.0693*** (−2.79)
Board size	0.781*** (10.57)	0.110*** (5.19)	0.177*** (17.83)	−0.00584 (−0.41)	−0.0573*** (−4.94)
CEO duality	0.0697** (2.09)	0.00198 (0.28)	0.0124*** (3.51)	−0.000452 (−0.10)	−0.0100*** (−2.64)
Board independence	1.091*** (10.33)	0.113*** (3.98)	0.265*** (19.80)	0.0105 (0.53)	−0.159*** (−10.01)
Female director	1.115*** (8.64)	−0.175*** (−4.69)	0.274*** (15.69)	−0.423*** (−16.52)	−0.0213 (−0.99)
N	20,038	19,896	19,896	19,896	19,896
Pseudo R-squared	0.229				
Adj R-squared		0.749	0.581	0.545	0.884
Year FE	No	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for robustness test for sample selection bias using Heckman two-stage method.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

TABLE 14 | Moderating effect of analyst coverage.

	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	−0.0253 (−1.12)	−0.00483 (−0.38)	−0.0227 (−1.55)	0.00182 (0.15)
Analyst	−0.0129*** (−3.42)	0.00000710 (0.00)	−0.00592** (−2.37)	−0.00754*** (−3.61)
SC×Analyst	0.0365*** (4.66)	−0.00193 (−0.44)	0.0298*** (5.92)	0.00873** (2.07)
CFO	−0.161** (−2.17)	−0.00365 (−0.12)	−0.105** (−2.29)	−0.0555 (−1.02)
Firm size	0.104*** (20.45)	−0.000825 (−0.36)	−0.00868*** (−2.58)	0.112*** (37.11)
Leverage	−0.00987 (−0.62)	0.0114* (1.67)	−0.00307 (−0.29)	−0.00986 (−0.99)
Sales growth	0.0149*** (3.37)	0.00175 (1.12)	0.0144*** (4.92)	−0.00272 (−0.92)
BTM	0.00205 (0.27)	0.00905*** (2.62)	−0.00746 (−1.54)	0.000872 (0.21)
Net PPE	−0.0281 (−0.81)	0.0171 (1.19)	−0.0483** (−2.18)	0.00520 (0.24)
Board size	−0.0402*** (−2.77)	−0.0120* (−1.79)	−0.0183* (−1.91)	−0.00814 (−0.99)
CEO duality	−0.00977 (−1.38)	−0.00256 (−0.72)	−0.00134 (−0.30)	−0.00606 (−1.64)
Board independence	−0.109*** (−6.26)	−0.00811 (−0.99)	−0.0113 (−0.96)	−0.0891*** (−8.75)
Female director	−0.426*** (−15.67)	−0.0344*** (−2.69)	−0.447*** (−23.85)	0.0566*** (3.58)
N	19,896	19,896	19,896	19,896
Adj. R-squared	0.747	0.560	0.546	0.884
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for moderation analysis on analyst coverage.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

When examining dimension-specific outcomes, the moderating effects of media scrutiny vary across ESG categories. For social decoupling (*DE_S*), the interaction term is positive and statistically significant (Panel A: $\beta = 0.127$, $p < 0.01$; Panel B: $\beta = 0.107$, $p < 0.01$), providing support for *Hypothesis 5b*. This indicates that media scrutiny amplifies the positive effect of SCs on social decoupling, suggesting that under high media visibility, firms with SCs are more likely to engage in externally oriented and

symbolically driven social initiatives that are easy to publicize. In contrast, the interaction term is negative and significant for governance decoupling (*DE_G*; Panel A: $\beta = -0.0539$, $p < 0.10$; Panel B: $\beta = -0.0573$, $p < 0.05$), which does not support *Hypothesis 5c*. This pattern suggests that media scrutiny may draw attention to governance weaknesses, prompting firms with SCs to reduce discrepancies between governance disclosures and actual practices. Media outlets may be more effective in identifying

TABLE 15 | Moderating effect of ESG-related media coverage.

Panel A: Yearly mean value of RRI				
	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.0373* (1.78)	0.0112 (0.93)	0.000895 (0.06)	0.0222** (2.07)
RRI_Mean	−0.0241 (−1.42)	−0.0138* (−1.68)	−0.0164 (−1.41)	0.00231 (0.24)
SC×RRI_Mean	0.0571 (0.97)	−0.0154 (−0.46)	0.127*** (3.36)	−0.0539* (−1.80)
CFO	−0.276** (−2.30)	−0.0998** (−2.00)	−0.141* (−1.94)	−0.0343 (−0.40)
Firm size	0.0862*** (12.05)	−0.00241 (−0.72)	−0.00860* (−1.85)	0.0968*** (23.85)
Leverage	−0.0169 (−0.77)	−0.00326 (−0.32)	−0.0107 (−0.73)	−0.00117 (−0.09)
Sales growth	0.00175 (0.25)	0.00671** (2.14)	0.00388 (0.79)	−0.00996** (−2.32)
BTM	0.0233** (2.27)	0.0100** (2.07)	0.00140 (0.21)	0.0129** (2.48)
Net PPE	0.00734 (0.17)	−0.00764 (−0.38)	−0.0262 (−0.90)	0.0391* (1.70)
Board size	−0.0150 (−0.80)	−0.00981 (−1.06)	−0.00400 (−0.32)	−0.00244 (−0.24)
CEO duality	−0.00747 (−0.89)	−0.00351 (−0.79)	0.00532 (1.00)	−0.0101** (−2.25)
Board independence	−0.101*** (−4.51)	−0.000246 (−0.02)	−0.0112 (−0.75)	−0.0899*** (−7.01)
Female director	−0.398*** (−11.53)	−0.0395** (−2.25)	−0.366*** (−15.72)	0.00647 (0.34)
N	11,298	11,298	11,298	11,298
Adj. R-squared	0.728	0.605	0.596	0.862
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Panel B: Yearly maximum value of RRI				
	(1) DE_CSR	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.0398* (1.85)	0.00945 (0.78)	0.00236 (0.17)	0.0253** (2.35)

(Continues)

TABLE 15 | (Continued)

Panel B: Yearly maximum value of RRI				
	(1) DE_CSR	(2) DE_E	(3) DE_S	(4) DE_G
RRI_Max	−0.0211 (−1.50)	−0.00994 (−1.47)	−0.0165* (−1.72)	0.00185 (0.23)
SC×RRI_Max	0.0422 (0.78)	−0.00804 (−0.27)	0.107*** (3.16)	−0.0573** (−2.08)
CFO	−0.277** (−2.31)	−0.100** (−2.01)	−0.141* (−1.94)	−0.0341 (−0.40)
Firm size	0.0862*** (12.06)	−0.00244 (−0.73)	−0.00859* (−1.85)	0.0967*** (23.84)
Leverage	−0.0167 (−0.76)	−0.00321 (−0.31)	−0.0106 (−0.72)	−0.00112 (−0.09)
Sales growth	0.00164 (0.24)	0.00675** (2.15)	0.00367 (0.75)	−0.00992** (−2.31)
BTM	0.0233** (2.27)	0.0100** (2.07)	0.00136 (0.20)	0.0130** (2.49)
Net PPE	0.00729 (0.17)	−0.00779 (−0.38)	−0.0263 (−0.90)	0.0393* (1.72)
Board size	−0.0151 (−0.81)	−0.00990 (−1.07)	−0.00424 (−0.34)	−0.00222 (−0.22)
CEO duality	−0.00745 (−0.89)	−0.00348 (−0.78)	0.00535 (1.00)	−0.0101** (−2.26)
Board independence	−0.101*** (−4.51)	−0.000324 (−0.03)	−0.0113 (−0.76)	−0.0898*** (−7.01)
Female director	−0.398*** (−11.53)	−0.0396** (−2.25)	−0.366*** (−15.70)	0.00618 (0.32)
N	11,298	11,298	11,298	11,298
Adj. R-squared	0.728	0.605	0.595	0.862
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for moderation analysis on ESG-related media coverage.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

governance-related inconsistencies, which can motivate firms to narrow governance decoupling. The interaction term for environmental decoupling (DE_E) is not statistically significant, indicating that there is no clear moderating effect of media scrutiny on environmental decoupling, and thus *Hypothesis 5a* is not supported. This implies that environmental outcomes, which typically require substantive operational changes, are less

sensitive to media scrutiny and cannot be easily influenced by external visibility alone.

Overall, ESG-related media scrutiny acts as a heterogeneous moderator. It strengthens social decoupling, slightly reduces governance decoupling, and has no significant effect on environmental decoupling among firms with SCs.

TABLE 16 | Moderating effect of institutional ownership.

	(1) DE_ESG	(2) DE_E	(3) DE_S	(4) DE_G
SC	0.162*** (3.56)	0.0469** (2.09)	0.0916*** (3.46)	0.0289 (1.42)
IO%	−0.0424* (−1.84)	0.0341*** (3.32)	−0.0731*** (−4.82)	−0.00633 (−0.45)
SC×IO%	−0.125** (−2.31)	−0.0736*** (−2.69)	−0.0583* (−1.80)	−0.00243 (−0.10)
CFO	−0.185** (−2.13)	−0.0408 (−1.16)	−0.121** (−2.39)	−0.0258 (−0.41)
Firm size	0.0927*** (16.60)	−0.00361 (−1.43)	−0.00477 (−1.32)	0.0999*** (30.94)
Leverage	−0.0289* (−1.69)	0.00468 (0.62)	−0.0106 (−0.94)	−0.0144 (−1.36)
Sales growth	0.0116** (2.48)	0.00271 (1.55)	0.00852*** (2.78)	−0.000769 (−0.24)
BTM	−0.000360 (−0.04)	0.00806** (2.14)	−0.00667 (−1.28)	−0.000958 (−0.22)
Net PPE	−0.0284 (−0.75)	0.0227 (1.43)	−0.0477** (−2.05)	−0.000126 (−0.01)
Board size	−0.0480*** (−3.12)	−0.00585 (−0.82)	−0.0214** (−2.13)	−0.0187** (−2.17)
CEO duality	−0.00761 (−1.02)	−0.00184 (−0.49)	−0.000339 (−0.07)	−0.00575 (−1.51)
Board independence	−0.124*** (−6.71)	−0.00805 (−0.92)	−0.0132 (−1.07)	−0.101*** (−9.49)
Female director	−0.420*** (−14.17)	−0.0273* (−1.92)	−0.421*** (−21.25)	0.0277 (1.62)
N	17,657	17,657	17,657	17,657
Adj. R-squared	0.755	0.563	0.568	0.891
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: This table reports the OLS regression results for *moderation analysis* on institutional ownership.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

4.8.3 | Moderating Effect of Institutional Ownership

The results reported in Table 16 show that institutional ownership significantly weakens the relationship between SCs and ESG decoupling. For overall ESG decoupling, the interaction term $SC \times IO\%$ is negative and statistically significant ($\beta = -0.125$, $p < 0.05$), supporting *Hypothesis 6*: Institutional ownership

negatively moderates the relationship between SCs and ESG decoupling. More specifically, under high levels of institutional ownership, firms with SCs exhibit a lower degree of ESG decoupling. Institutional investors often exert pressure for substantive ESG integration, which can constrain firms from engaging in symbolic or impression-management practices. As a result, SCs may be less able to rely on superficial disclosure strategies when ownership is

concentrated among investors who demand credible and meaningful ESG performance. Consequently, institutional ownership dampens the positive association between SCs and ESG decoupling, indicating a shift away from externally oriented signaling and toward more genuine sustainability integration.

Examining dimension-specific outcomes, the interaction term $SC \times IO\%$ is negative and statistically significant for environmental decoupling (DE_E ; $\beta = -0.0736$, $p < 0.01$) and social decoupling (DE_S ; $\beta = -0.0583$, $p < 0.01$), providing support for *Hypotheses 6a* and *6b*. These findings indicate that institutional ownership weakens the positive effect of SCs on environmental and social decoupling, suggesting that under strong institutional monitoring, firms with SCs are less likely to engage in symbolic ESG practices in these dimensions. Institutional investors appear to exert effective monitoring and disciplinary pressure, encouraging more substantive alignment between ESG disclosures and actual practices.

In contrast, the interaction term for governance decoupling (DE_G) is statistically insignificant, indicating that there is no moderating effect of institutional ownership on governance decoupling, and thus *Hypothesis 6c* is not supported. This suggests that governance outcomes are less sensitive to ownership-based oversight, potentially limiting the ability of institutional investors to influence governance-related decoupling.

Overall, the findings support *Hypothesis 6* and its sub-hypotheses (*Hypotheses 6a* and *6b*), but not *Hypothesis 6c*, highlighting that institutional ownership plays a disciplining role by reducing ESG decoupling, particularly in environmental and social dimensions, thereby constraining symbolic practices and promoting more substantive ESG alignment.

5 | Conclusions and Implications

5.1 | Conclusions

In this study, we find that SCs are associated with higher overall ESG decoupling, suggesting that firms may strategically use these committees to expand external sustainability disclosures without making corresponding improvements in internal ESG practices. Although SCs are designed as governance mechanisms intended to strengthen oversight and accountability, they may unintentionally encourage symbolic sustainability behavior by prioritizing visible ESG initiatives that enhance legitimacy and public image rather than ensuring alignment between internal practices and external claims. We also find that the influence of SCs varies across ESG dimensions. Specifically, SCs reduce environmental decoupling but increase social and governance decoupling. These findings suggest that although SCs contribute to genuine improvements in environmental performance, they simultaneously reinforce symbolic ESG behavior in the social and governance domains.

Analysis of internal versus external ESG actions clarifies these patterns. SCs significantly increase external ESG initiatives but do not affect overall internal ESG activities, indicating that firms tend to focus on visible, legitimacy-oriented actions rather than substantive internal reforms. Dimension-specific results show that SCs support meaningful improvements in both internal and

external environmental actions, which reduces environmental decoupling. In contrast, SCs enhance external social and governance initiatives while exerting limited influence on internal practices in these areas. This imbalance contributes to symbolic reporting behavior and heightened decoupling in the social and governance dimensions.

Finally, the moderating analyses identify conditions under which these dynamics intensify or weaken. Analyst coverage strengthens the positive association between SCs and overall ESG decoupling, suggesting that heightened scrutiny from financial analysts may paradoxically encourage firms to expand symbolic practices by increasing external communication efforts without advancing internal ESG performance. Dimension-specific results show that analyst coverage increases social decoupling, indicating that under strong analyst attention, SCs are more likely to emphasize social initiatives externally even when internal adoption remains limited. Analyst coverage also increases governance decoupling, suggesting that SCs may focus on enhancing outward-facing governance policies and disclosures to satisfy analysts, even if internal governance systems do not substantially improve. In contrast, analyst coverage has no significant effect on environmental decoupling, implying that environmental disclosures tend to remain more closely aligned with internal environmental practices, even under intensive scrutiny.

Media scrutiny does not moderate the relationship between SCs and overall ESG decoupling, indicating that general media attention does not alter how SCs influence the gap between ESG disclosures and actual performance. When the ESG dimensions are examined separately, however, media scrutiny produces differentiated effects. Media attention increases social decoupling, suggesting that under heightened media visibility, firms expand symbolic social initiatives without making proportionate internal improvements. At the same time, media scrutiny slightly reduces governance decoupling, implying that public exposure may prompt firms to strengthen internal governance structures and compliance mechanisms, which narrows the gap between governance disclosures and internal practices. Media scrutiny shows no moderating effect on environmental decoupling, indicating that environmental outcomes remain largely unaffected by variation in media attention.

Institutional ownership weakens the positive relationship between SCs and overall ESG decoupling, suggesting that a greater presence of institutional investors may limit symbolic ESG practices and encourage firms to better align external disclosures with internal ESG activities. When the ESG dimensions are analyzed separately, institutional ownership again shows differentiated effects. It reduces environmental decoupling, indicating that institutional investors promote substantive improvements in environmental performance and help ensure that external reporting reflects internal actions. Institutional ownership also decreases social decoupling, suggesting that investors press firms to prioritize meaningful social initiatives rather than rely on outward-facing communication alone. In contrast, institutional ownership shows no significant moderating effect on governance decoupling, implying that the alignment between governance disclosures and internal governance practices is largely unaffected by investor presence.

Overall, these findings indicate that SCs primarily facilitate symbolic ESG engagement rather than substantive transformation, except in the environmental domain. The results highlight the importance of both internal governance mechanisms and external monitoring in shaping ESG outcomes and provide insights for regulators, investors, and corporate boards seeking to reduce ESG decoupling and strengthen credible sustainability practices.

5.2 | Theoretical Contributions

This study makes several theoretical contributions to the literature. First, prior research has often examined ESG performance and disclosure using frameworks such as agency theory (Jensen and Meckling 1976), legitimacy theory (Suchman 1995), and stakeholder theory (Freeman 1984). However, to our knowledge, this study is the first to examine the relationship between board-level SCs and ESG decoupling within an integrated agency-legitimacy-impression management framework (Goffman 1959). Although SCs are designed to improve oversight and accountability, our findings reveal that they can simultaneously encourage symbolic ESG behavior, where external disclosures are enhanced without corresponding internal improvements, showing that governance mechanisms can promote both substantive and symbolic ESG actions.

Second, this study extends theory by examining dimension-specific ESG decoupling. SCs reduce environmental decoupling (DE_E) but increase social (DE_S) and governance decoupling (DE_G). These results highlight that SCs do not affect all ESG dimensions uniformly, extending agency and legitimacy theories by demonstrating how board mechanisms influence substantive versus symbolic behavior differently across ESG domains.

Third, by differentiating internal versus external ESG actions, the study provides new insights into impression management processes. SCs significantly increase external ESG actions (e.g., disclosures, campaigns, and reporting) while having limited effects on internal ESG practices. This aligns with impression management theory (Goffman 1959) and expands our understanding of how firms strategically allocate ESG efforts to enhance legitimacy rather than drive internal sustainability improvements.

Fourth, the study contributes to the literature on external monitoring by showing that moderating factors such as analyst coverage, media scrutiny, and institutional ownership influence the SC–ESG decoupling relationship. For example, analyst coverage amplifies social and governance decoupling, whereas institutional ownership reduces decoupling in environmental and social domains. These findings extend institutional and legitimacy theories by highlighting how external pressures interact with internal governance to shape ESG outcomes.

Fifth, the study advances theory by integrating overall ESG decoupling, dimension-specific effects, internal/external actions, and moderating conditions into a comprehensive framework. This approach provides a nuanced understanding of when and how SCs promote symbolic versus substantive ESG behavior,

offering a foundation for future research on board-level sustainability governance across contexts, industries, and institutional settings.

5.3 | Practical Implications

This study has practical implications for regulators, policymakers, ESG rating agencies, investors, and corporate leaders. First, the results indicate that SCs can inadvertently increase symbolic ESG behavior, particularly in the social and governance dimensions, while contributing to substantive improvements in environmental actions. Regulators and standard-setting bodies should strengthen ESG reporting frameworks by mandating standardized, verifiable, and dimension-specific disclosures. The voluntary and unstandardized nature of current ESG reporting allows firms to prioritize visibility over internal effectiveness. Any mandatory regulations should emphasize consistency, comparability, robust assurance mechanisms, and penalties for misreporting.

Second, the distinction between internal and external ESG actions highlights that firms often prioritize visible, outward-facing initiatives rather than meaningful internal practices. Corporate boards and policymakers should design mechanisms to promote genuine internal ESG improvements, such as integrating SC oversight with executive evaluations and linking compensation to actual ESG performance rather than disclosure metrics alone.

Third, the study finds that external monitoring factors influence ESG decoupling differently across dimensions. Analyst coverage tends to increase symbolic social and governance initiatives, whereas institutional ownership reduces decoupling in environmental and social domains. This suggests that investors, analysts, and boards should consider both internal ESG performance and external disclosures to evaluate credibility and avoid misleading signals.

Fourth, SCs are most effective in reducing environmental decoupling, suggesting that environmental practices are more closely aligned with internal actions. Corporate leaders can leverage this by implementing structured environmental governance systems, such as dedicated reporting teams, ESG dashboards, or formalized environmental committees, to enhance transparency and reduce the risk of symbolic behavior.

Fifth, ESG rating agencies should focus on dimension-specific monitoring and differentiate between internal and external ESG activities. Improving the granularity, accuracy, and contextualization of ESG scores would help investors and regulators identify symbolic actions, reward substantive improvements, and incentivize authentic sustainability practices.

Sixth, the study shows that relying solely on market forces, such as analyst scrutiny or investor pressure, may be insufficient to prevent ESG decoupling. Regulators should develop proactive early-warning systems incorporating AI-driven ESG monitoring tools, advanced data analytics, and robust whistleblower mechanisms to detect symbolic ESG behavior early and maintain market integrity.

Coordinated action involving regulatory reform, enhanced corporate governance, investor oversight, and robust ESG verification is essential to promote transparent and credible sustainability practices.

5.4 | Limitations and Future Research Directions

This study has several limitations that future research could address. First, the analysis focuses on publicly listed firms in a single country. Future studies should examine other regions, industries, and firm types to assess the generalizability of SC effects on ESG decoupling. Second, our study relies on secondary ESG and financial data, which may not capture the full range of symbolic or substantive ESG initiatives. Incorporating survey-based or qualitative data, such as interviews with managers, could provide richer insights into the decision-making processes behind SCs and ESG strategies. Third, although we consider moderators such as analyst coverage, institutional ownership, and media scrutiny, other firm-level or contextual factors may influence ESG decoupling. Future research could explore board characteristics, ownership concentration, corporate culture, and stakeholder engagement to develop a more complete understanding of ESG governance. Last, we focus primarily on ESG decoupling outcomes. Future studies could examine the operational, reputational, and financial consequences of symbolic versus substantive ESG actions to understand the broader impact of SC-driven sustainability initiatives.

Author Contributions

Weite Qiu and Maqsood Ahmad contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Weite Qiu, Yuming Xuan, and Maqsood Ahmad. Yuming Xuan contributed to the early stage of the research by assisting in data collection and variables operationalization but did not continue in the later analytical phase. She was re-invited during the revision stage and made major contributions by incorporating reviewers' comments, refining the empirical analysis, strengthening the robustness checks, and improving the discussion section. The first draft of the manuscript was written by Weite Qiu and Maqsood Ahmad. Sunny Sun and Yang Jinghan contributed by supporting theoretical development, data interpretation, and critically reviewing the initial draft. All authors have read and approved the final manuscript and agree to be accountable for all aspects of the work.

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Appendix A

Variable Definitions

Variable name	Definition/measurement
Independent variable	
SC	An indicator variable coded as 1 if a firm has a sustainability committee, 0 otherwise, identified from BoardEx;
Dependent variables	
DE_ESG	The overall ESG decoupling that equals the difference between the claims/disclosure and policies/practice of a firm related to ESG and then scaled by the natural logarithm of total assets, data for a firm's ESG claims and policies is collected from Refinitiv ESG;
DE_E	The environmental decoupling that equals the difference between the claims/disclosure and policies/practice of a firm related to environmental issues and then scaled by the natural logarithm of total assets, from Refinitiv ESG;
DE_S	The social decoupling that equals the difference between the claims/disclosure and policies/practice of a firm related to social issues and then scaled by the natural logarithm of total assets, from Refinitiv ESG;
DE_G	The governance decoupling that equals the difference between the claims/disclosure and policies/practice of a firm related to corporate governance and then scaled by the natural logarithm of total assets, from Refinitiv ESG;
External	The sum of a firm's claims or disclosure on ESG scaled by the natural logarithm of total assets, from Refinitiv ESG;
Internal	The sum of a firm's policies or practice on ESG scaled by the natural logarithm of total assets, from Refinitiv ESG;
EX_E	The sum of a firm's claims or disclosure on environmental issues scaled by the natural logarithm of total assets, from Refinitiv ESG;
EX_S	The sum of a firm's claims or disclosure on social issues scaled by the natural logarithm of total assets, from Refinitiv ESG;
EX_G	The sum of a firm's claims or disclosure on corporate governance scaled by the natural logarithm of total assets, from Refinitiv ESG;
IN_E	The sum of a firm's policies or practice on environmental issues scaled by the natural logarithm of total assets, from Refinitiv ESG;
IN_S	The sum of a firm's policies or practice on social issues scaled by the natural logarithm of total assets, from Refinitiv ESG;
IN_G	The sum of a firm's policies or practice on corporate governance scaled by the natural logarithm of total assets, from Refinitiv ESG;
Moderating variables	
Analyst	The natural logarithm of 1 plus the number of analyst forecasts on earnings, measured at fiscal year-end, from IBES;
IO%	The ratio of total institutional ownership in the percentage of market value, from FactSet;
RRI_Mean	The natural logarithm of 1 plus the yearly mean value of RepRisk Index, scaled by the natural logarithm of total assets, from RepRisk;
RRI_Max	The natural logarithm of 1 plus the yearly mean value of RepRisk Index, scaled by the natural logarithm of total assets, from RepRisk;
Control variables	
ESG score	A variable that ranges from 0 to 1 and measures a firm's ESG performance based on verifiable public information, from Refinitiv ESG;
Ind_Mean ESG	The yearly industrial mean value of the ESG Score, from Refinitiv ESG;
Envrn score	A variable that measures a firm's environmental performance, from Refinitiv ESG;
Social score	A variable that measures a firm's social performance, from Refinitiv ESG;
Corpgov score	A variable that measures a firm's corporate governance, from Refinitiv ESG;

Variable name	Definition/measurement
Ind_Envrn	An indicator variable coded as 1 if a firm's two-digit historical SIC code is between 10 and 39, 0 otherwise, from Compustat;
Firm size	The natural logarithm of total assets, measured at fiscal year-end, from Compustat;
CFO	Cash flows from operation scaled by total assets, measured at fiscal year-end, from Compustat;
Leverage	The ratio of total liabilities to total assets, measured at fiscal year-end, from Compustat;
Sales growth	The percentage change of sales from the previous year to current year, from Compustat;
BTM	The ratio of book value of equity to the market value of equity, measured at fiscal year-end, from Compustat;
Net PPE	Net balance of property, plant, and equipment scaled by total assets, measured at fiscal year-end, from Compustat;
Board size	The natural logarithm of the number of directors sitting on the board, from BoardEx;
CEO duality	An indicative variable coded as 1 if the CEO of a firm also serve as the chairman of the board, 0 otherwise, from BoardEx;
Board independence	The ratio of the number of independent directors to the total number of directors, from BoardEx;
Female director	The ratio of the number of female directors to the total number of directors, from BoardEx;